



Sri Lanka Institute of Information Technology
IE2012: System and Network Programming
Year 2 Semester 1

Assignment

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1. Basics of the Linux environment

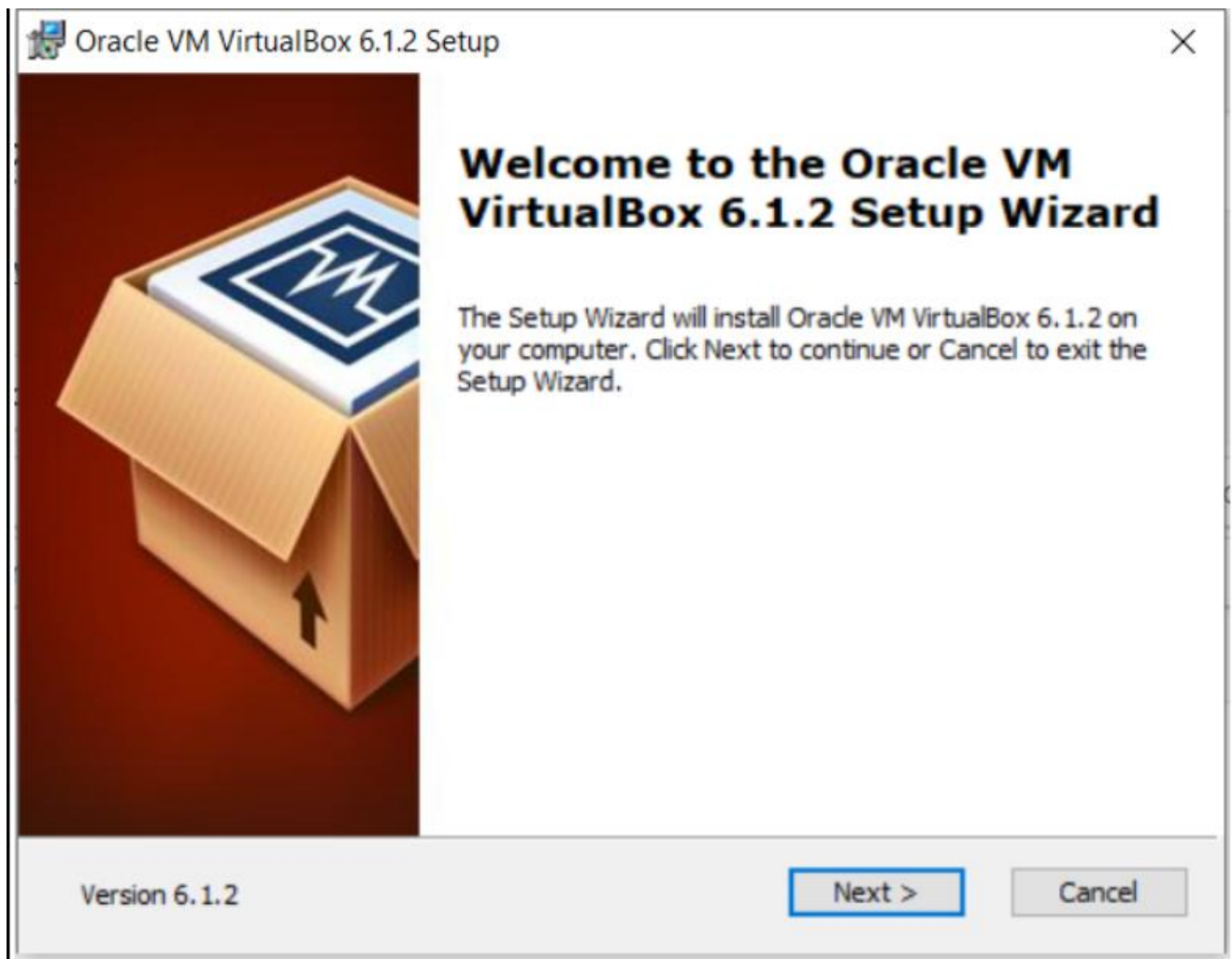
I. Virtual Machine Setup

Step 01

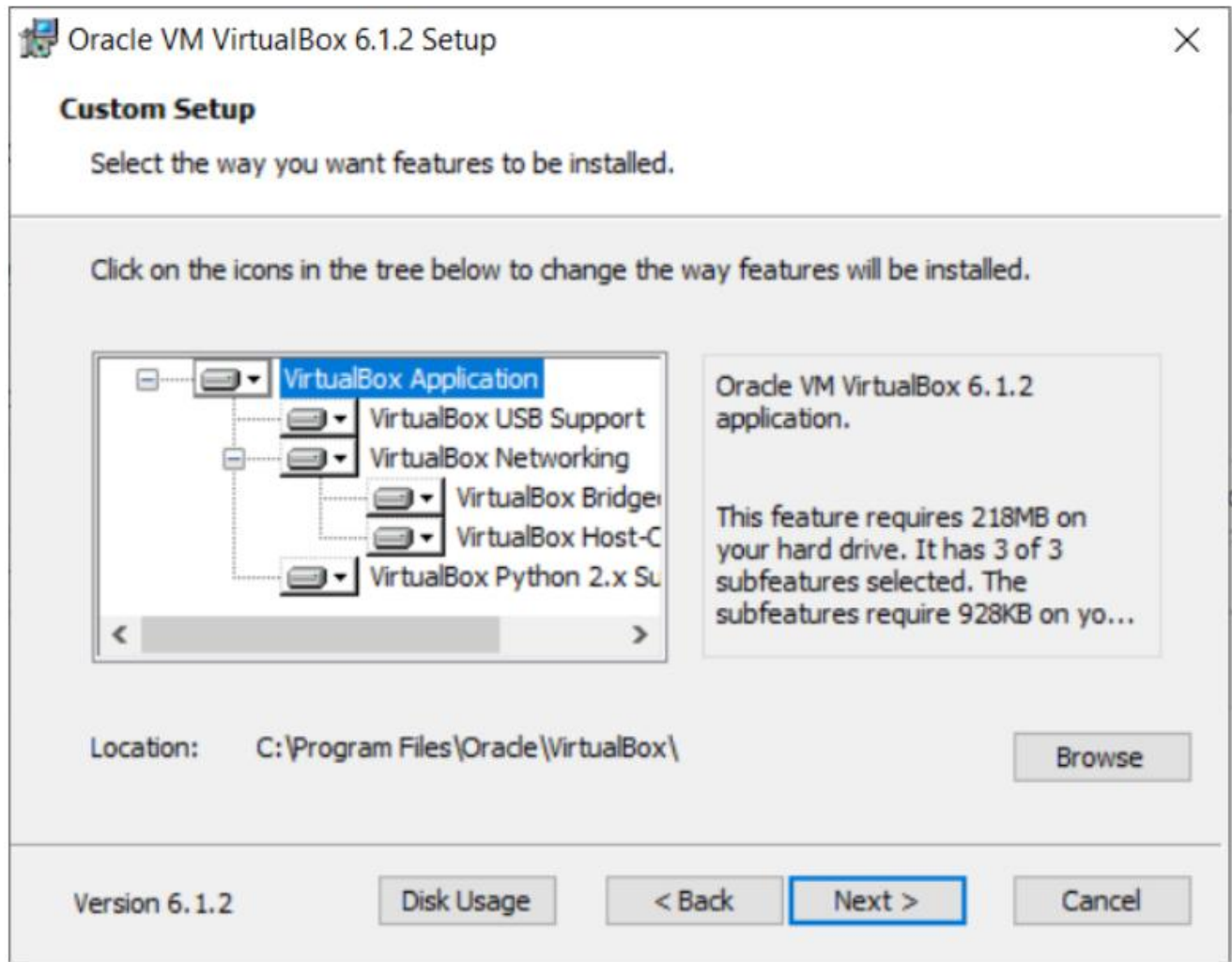
Download a virtual emulator for Windows, Linux, or Mac to start. To download VirtualBox, go to the official site [virtualbox.org](https://www.virtualbox.org) and download the latest version for windows.



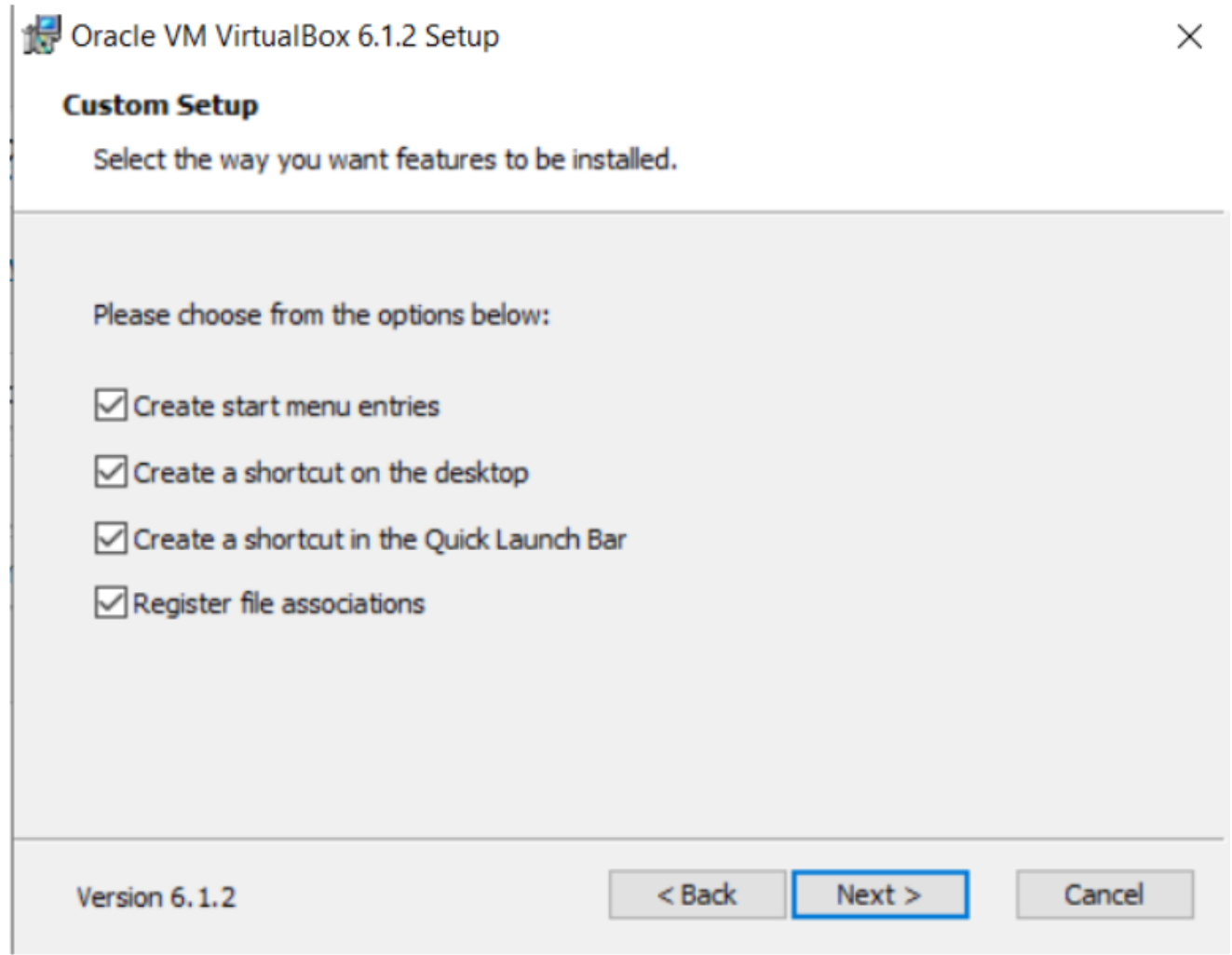
- **Getting Started:**



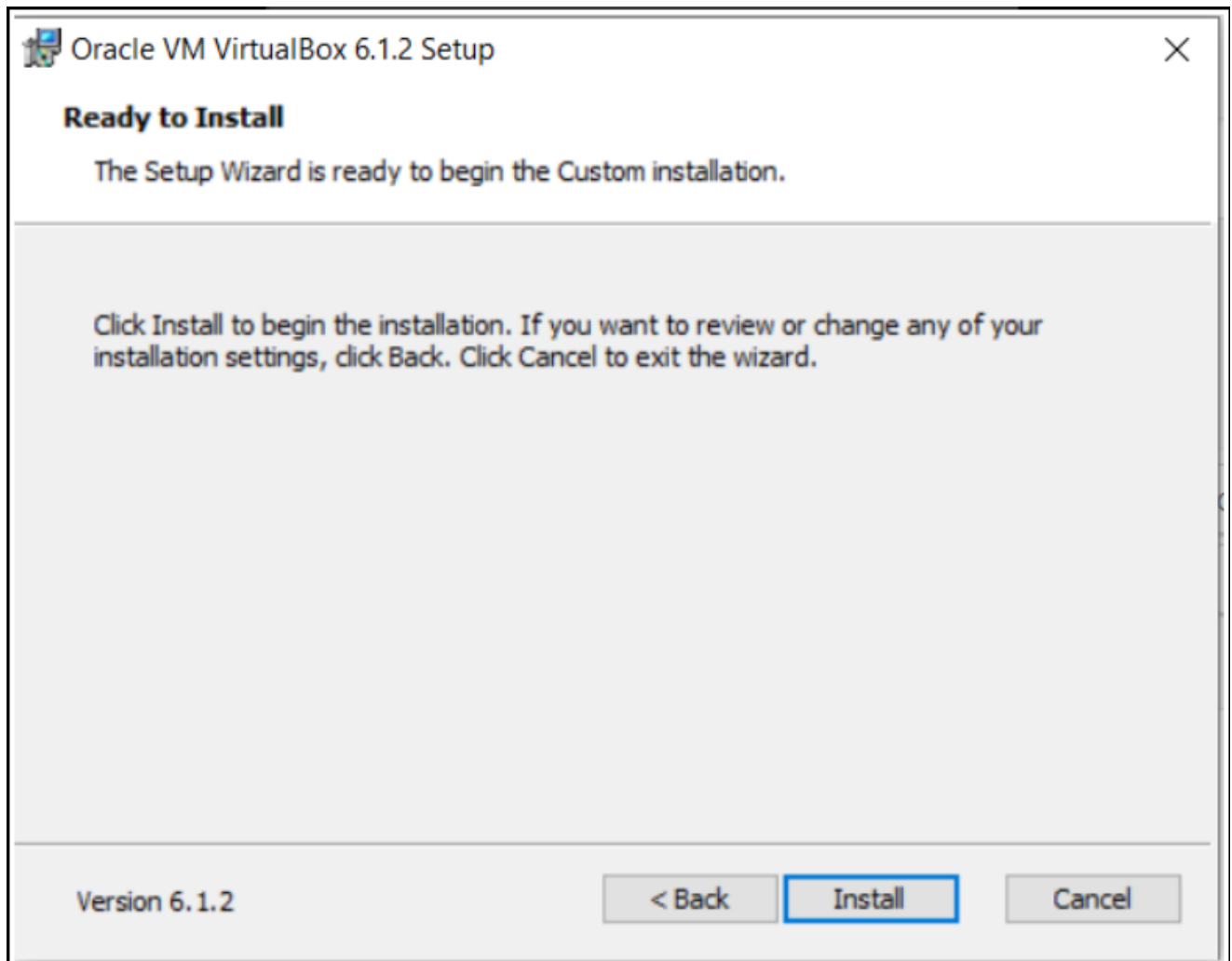
- **Select Installation Location:**



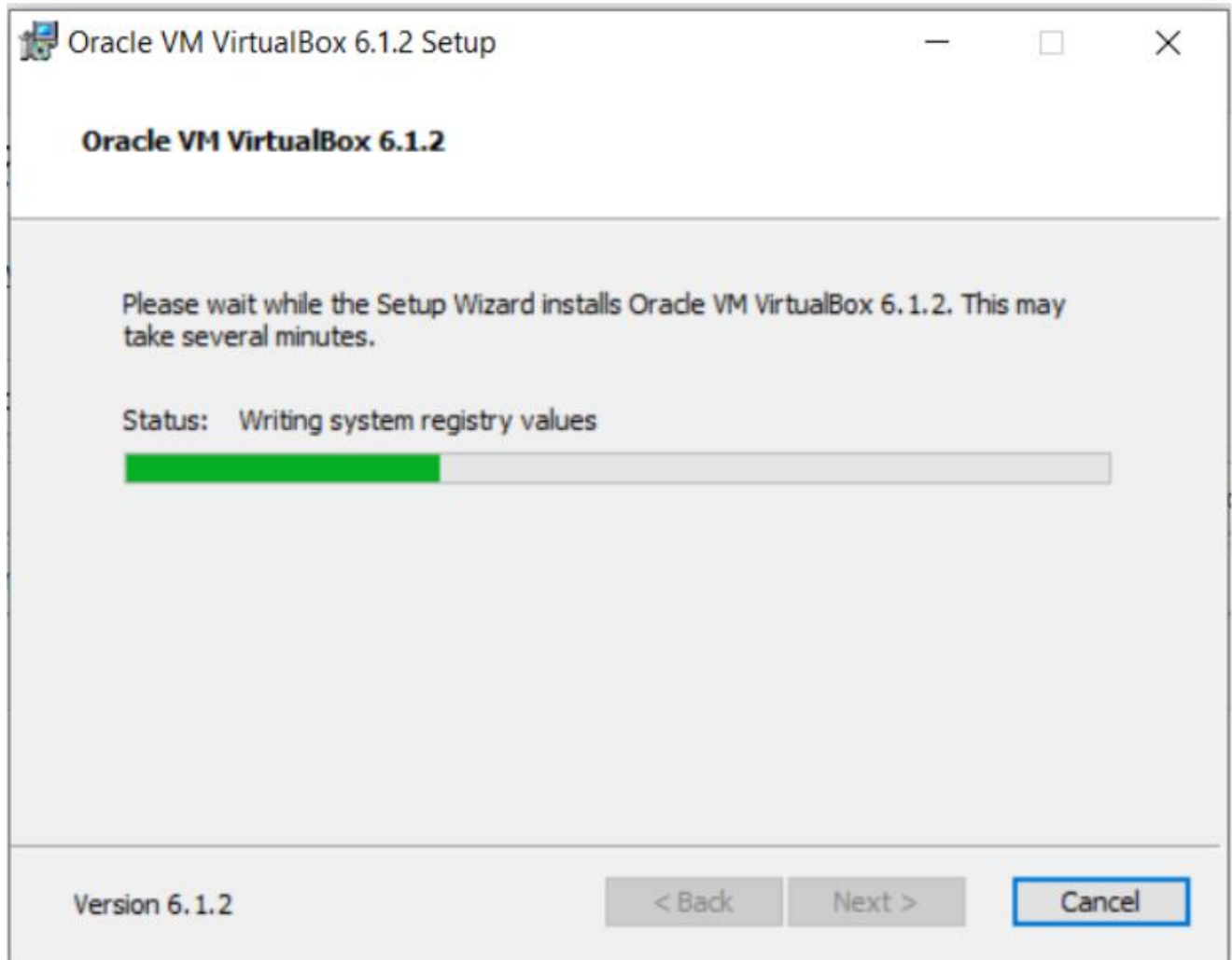
- **Creating Entries and Shortcuts:**



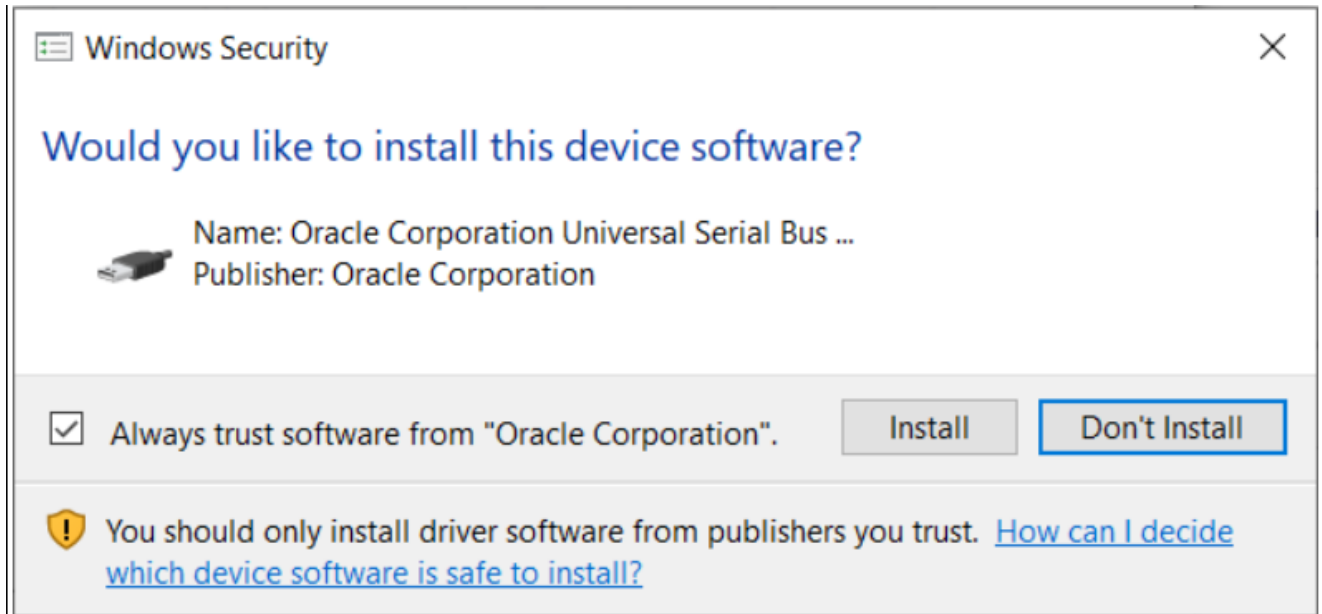
- **Ready to Install:**



- Installing Files and packages:



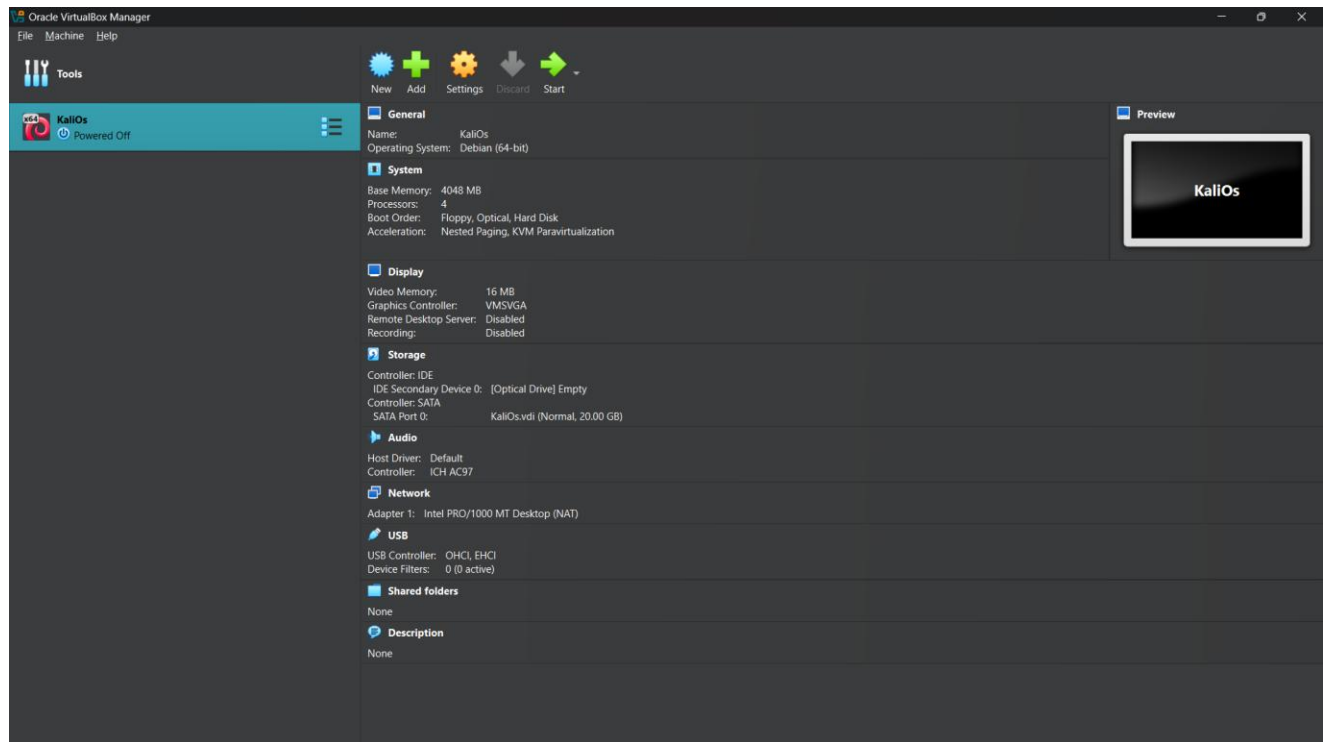
- **Installing Certificates:**



- **Finished Installation:**



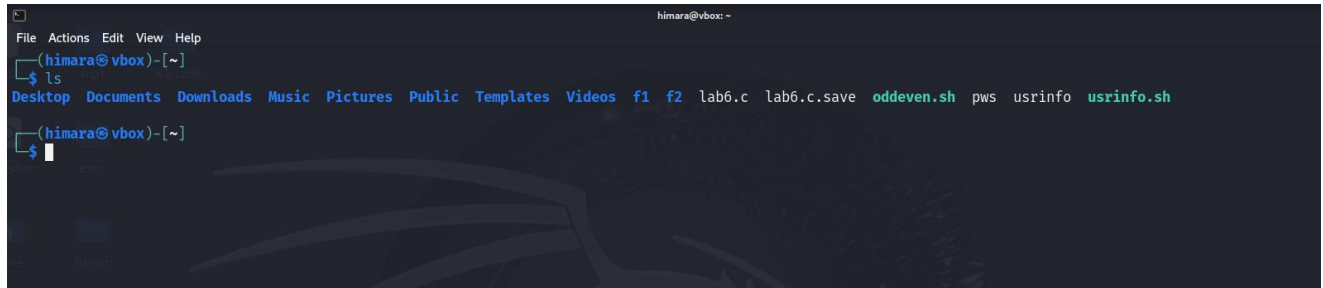
We can now continue with the installation (ensure you have Microsoft Visual C++ 2019 redistribution or higher version installed). Once the executable file is run and the administrator privilege supplement pop-up is allowed, we should choose the location and mode of Virtual Box installation. This box will appear when we start Virtual Box via the new desktop shortcut icon once the installation process has finished.



We can select the “New” option in the upper left corner and enter the information for the VM. We need to provide it a name, install location, and installation media image file in order of precedence. The other fields will also be filled automatically with the selected ISO file as soon as you select it. The Linux operating system will need to be installed manually if the check box "Skip Unattended Installation" is selected; otherwise, it will install automatically without our intervention. We can then select the next button by clicking.

II. Command Line Introduction

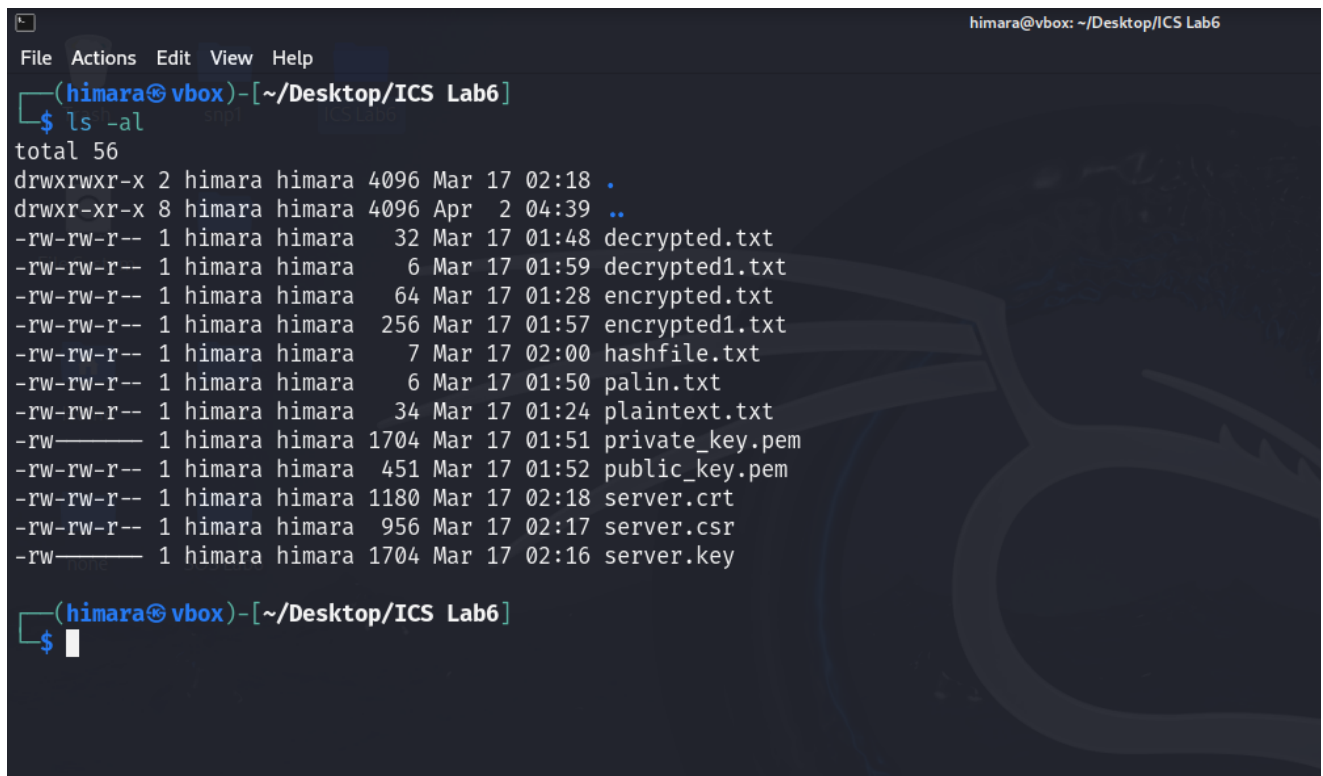
1. **ls – Listing files:** This command is used to list the files and directories stored in the current directory. When you type ls and press Enter, the current directory's contents may be viewed as a list.



```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos f1 f2 lab6.c lab6.c.save oddeven.sh pws usrinfo usrinfo.sh  
(himara@vbox)-[~]  
$
```

Other options available:

- a) **-l** : Long format which shows file details like permissions, size and modification time.
- b) **-a** : Lists all files, including hidden ones (files that start with a dot).



```
himara@vbox: ~/Desktop/ICS Lab6  
File Actions Edit View Help  
(himara@vbox)-[~/Desktop/ICS Lab6]  
$ ls -al  
total 56  
drwxrwxr-x 2 himara himara 4096 Mar 17 02:18 .  
drwxr-xr-x 8 himara himara 4096 Apr  2 04:39 ..  
-rw-rw-r-- 1 himara himara  32 Mar 17 01:48 decrypted.txt  
-rw-rw-r-- 1 himara himara   6 Mar 17 01:59 decrypted1.txt  
-rw-rw-r-- 1 himara himara  64 Mar 17 01:28 encrypted.txt  
-rw-rw-r-- 1 himara himara 256 Mar 17 01:57 encrypted1.txt  
-rw-rw-r-- 1 himara himara   7 Mar 17 02:00 hashfile.txt  
-rw-rw-r-- 1 himara himara   6 Mar 17 01:50 palin.txt  
-rw-rw-r-- 1 himara himara  34 Mar 17 01:24 plaintext.txt  
-rw----- 1 himara himara 1704 Mar 17 01:51 private_key.pem  
-rw-rw-r-- 1 himara himara  451 Mar 17 01:52 public_key.pem  
-rw-rw-r-- 1 himara himara 1180 Mar 17 02:18 server.crt  
-rw-rw-r-- 1 himara himara  956 Mar 17 02:17 server.csr  
-rw----- 1 himara himara 1704 Mar 17 02:16 server.key  
(himara@vbox)-[~/Desktop/ICS Lab6]  
$
```

- You have, from left to right:

- the file permissions (and if your system supports ACLs, you get an ACL flag as well)
- the number of links to that file
- the owner of the file
- the group of the file
- the file size in bytes
- the file modified datetime
- the file name

2. **cd – Change directory:** This Using the command-line interface, it is used to modify the current working directory. If you want to get into a directory called "Downloads" for instance, type `cd Downloads`.

```
himara@vbox: ~/Downloads
File Actions Edit View Help
(himara@vbox)-[~]
$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos f1 f2 lab6.c lab6.c.save oddeven.sh pws usrinfo usrinfo.sh
(himara@vbox)-[~]
$ cd Downloads
(himara@vbox)-[~/Downloads]
$
```

To move back a directory, we can use: **`..cd`**

- Examples:

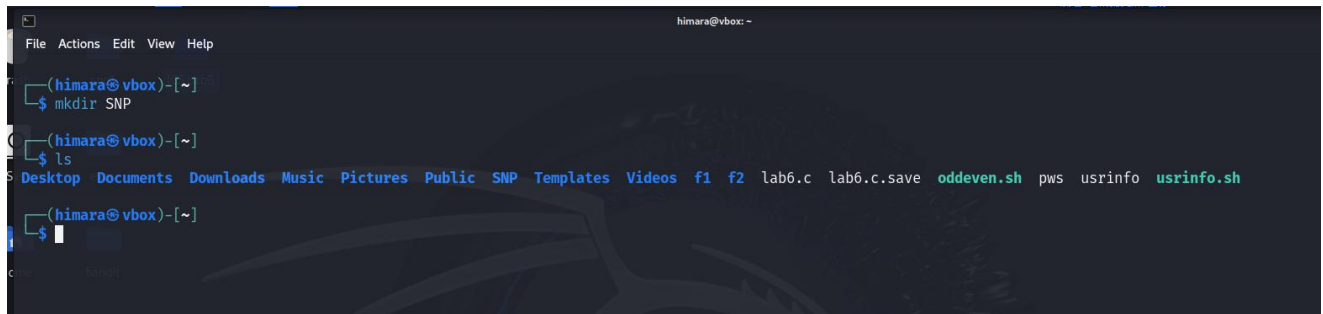
`cd /home/user/Downloads`: to move to the Downloads directory.

`cd ..` :to move up one directory level.

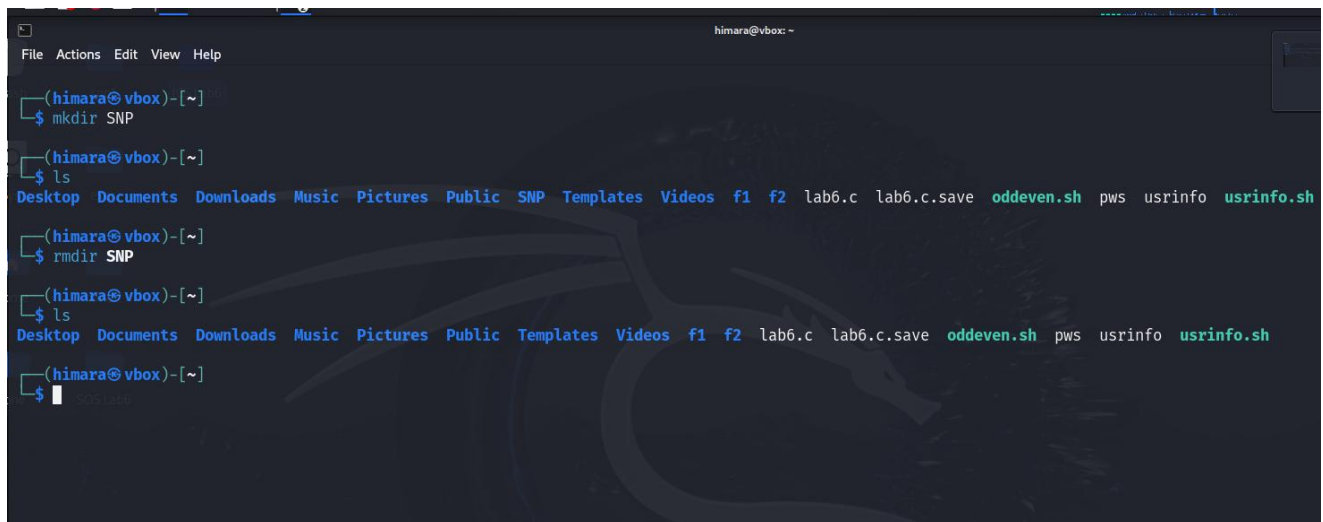
3. **pwd – present working directory:** Show your current full path that you in and it helps to you know where you are in the file system.

```
(himara@vbox)-[~]
$ pwd
/home/himara
```

4. **mkdir – Make Directory:** This command use to create new folders or directories

A terminal window titled 'himara@vbox: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(himara@vbox)-[~]'. The user enters '\$ mkdir SNP'. The prompt changes to '(himara@vbox)-[~]'. The user enters '\$ ls'. The terminal shows a directory listing: Desktop, Documents, Downloads, Music, Pictures, Public, SNP, Templates, Videos, f1, f2, lab6.c, lab6.c.save, oddeven.sh, pws, usrinfo, and usrinfo.sh. The prompt is now '\$ '.

5. **rmdir – Remove Directory:** This command is used to remove folders or directories, but it can only remove them if they are empty.

A terminal window titled 'himara@vbox: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(himara@vbox)-[~]'. The user enters '\$ mkdir SNP'. The prompt changes to '(himara@vbox)-[~]'. The user enters '\$ ls'. The terminal shows a directory listing: Desktop, Documents, Downloads, Music, Pictures, Public, SNP, Templates, Videos, f1, f2, lab6.c, lab6.c.save, oddeven.sh, pws, usrinfo, and usrinfo.sh. The prompt is now '\$ '. The user enters '\$ rmdir SNP'. The prompt changes to '(himara@vbox)-[~]'. The user enters '\$ ls'. The terminal shows a directory listing: Desktop, Documents, Downloads, Music, Pictures, Public, Templates, Videos, f1, f2, lab6.c, lab6.c.save, oddeven.sh, pws, usrinfo, and usrinfo.sh. The prompt is now '\$ '.

6. **rm -r:** If the folder or directory is not empty you can use this command. Be careful as this command does not ask for confirmation, and it will immediately remove anything you ask it to remove. There is no bin when removing files from the command line, and recovering lost files can be hard
- ☐ rm → remove
 - ☐ -r → recursively delete **everything inside** the folder

7. **cp – Copy files/folders:** Copy the contents of source file to the destination file called destination-file. For example, to copy a file named "test. txt" from a folder "SNP" to a file named "test1. txt".

```
(himara@vbox)-[~]
$ cat text.txt
Hiii SNP!!!

(himara@vbox)-[~]
$ cat text1.txt

(himara@vbox)-[~]
$ cp text.txt SNP/text1.txt

(himara@vbox)-[~]
$ cd SNP

(himara@vbox)-[~/SNP]
$ cat text1.txt
Hiii SNP!!!

(himara@vbox)-[~/SNP]
```

8. **mv – Move or Rename:** This command is used to move files/directories from one location to another, or to rename them. Simple syntax is **mv source_file destination_file to move a file.**

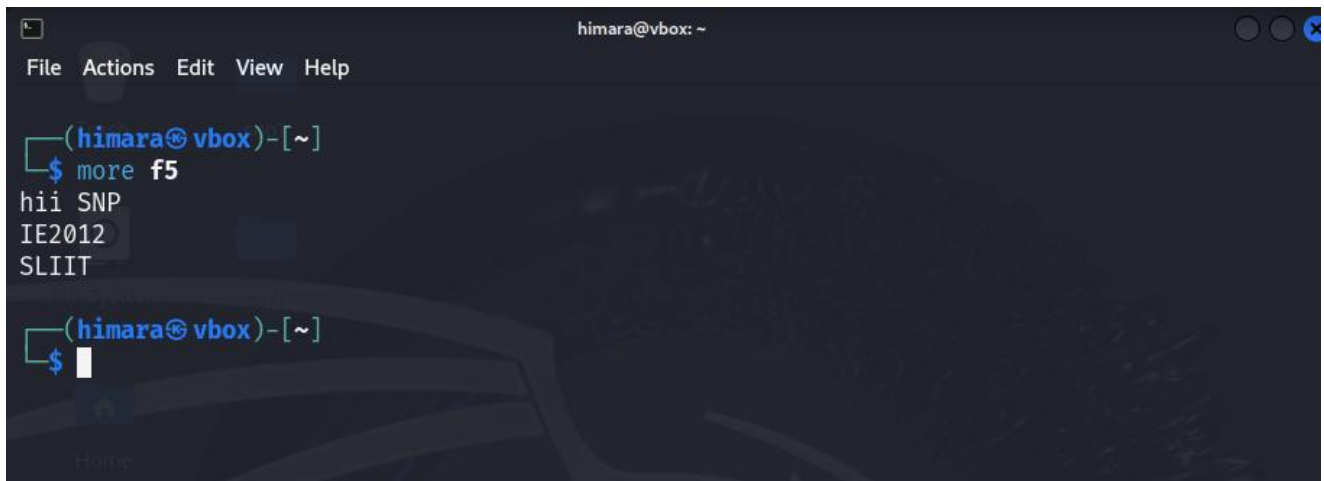
As an example, if we want to transfer a file called "test. txt" to a directory named "SNP" you would use: test1.txt

SNP/test2.txt . To change a name to a file, you would use mv old_filename new_filename.

```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ cat > f5  
himara@vbox: ~/SNP  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ ls  
Desktop      Music      SNP      f1      lab6.c.save  text.txt  usrinfo.sh  
Documents    Pictures   Templates f2      oddeven.sh  text1.txt  
Downloads    Public    Videos  lab6.c  pws         usrinfo  
(himara@vbox)-[~]  
$ cd SNP  
(himara@vbox)-[~/SNP]  
$ ls  
test.txt  test1.txt  text.txt  text1.txt  
(himara@vbox)-[~/SNP]  
$ cd ..  
(himara@vbox)-[~]  
$ mv text.txt SNP/text2.txt  
(himara@vbox)-[~]  
$ cd SNP  
Saved to this PC  
(himara@vbox)-[~/SNP]  
$ ls  
test.txt  test1.txt  text.txt  text1.txt  text2.txt  
(himara@vbox)-[~/SNP]  
$ cat text2.txt  
Hiii SNP!!!  
(himara@vbox)-[~/SNP]  
$
```

9. **cat – Concatenate and Display:** This is used for display or concatenate the content of a file. If it is an ASCII text file it will show everything inside that file.

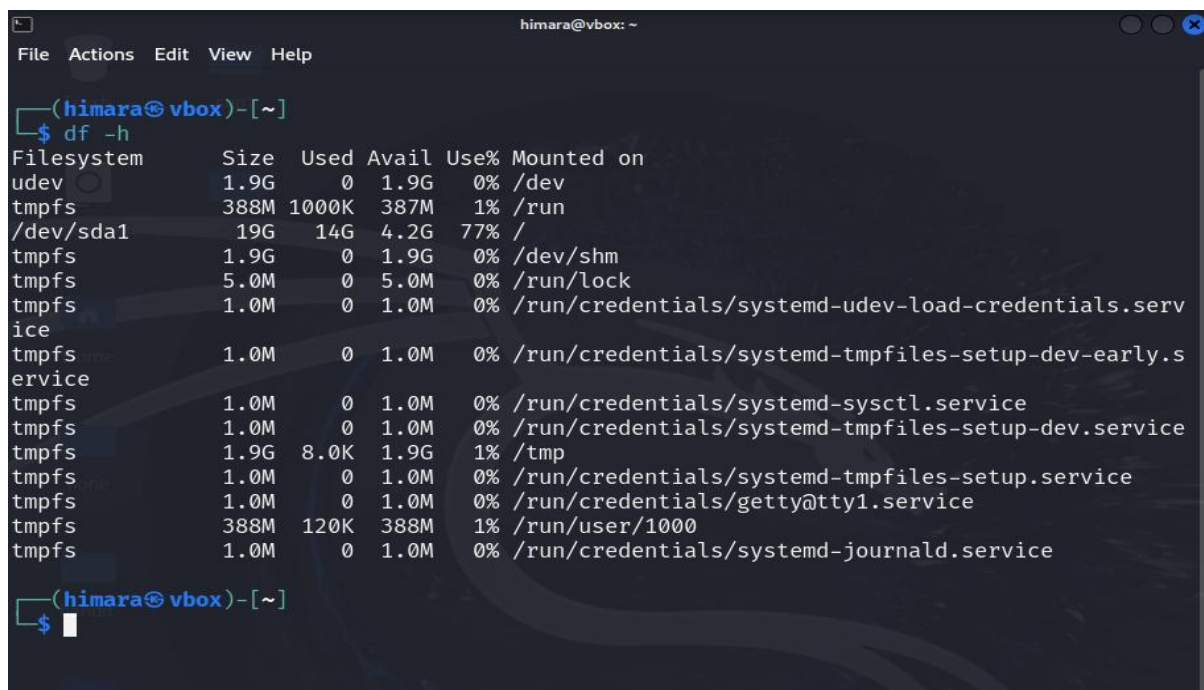
10. **more:** This command shows a simple pager program that is used to display the text of a text file page by page. It allows one to read large files in a manageable fashion by scrolling page by page.



```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ more f5  
hii SNP  
IE2012  
SLIIT  
(himara@vbox)-[~]  
$
```

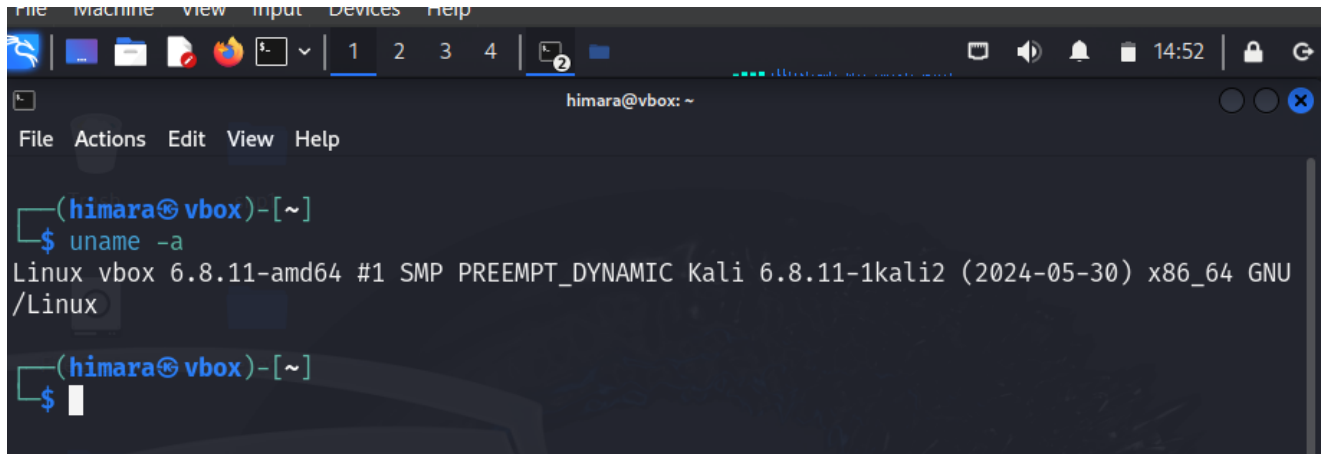
III. System Information and User Management

1. **df – Disk free:** Shows disk usage in human-readable format (GB/MB). This command helps track how full your partitions are. We can use the “-h” flag to show the size information displayed in units of 1024 i. e. k, M, G and T. The default will be showing the sizes in Kilobytes if the “-h” flag is not added.



```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
udev            1.9G   0    1.9G   0% /dev  
tmpfs           388M 1000K 387M   1% /run  
/dev/sda1       19G   14G  4.2G  77% /  
tmpfs           1.9G   0    1.9G   0% /dev/shm  
tmpfs           5.0M   0    5.0M   0% /run/lock  
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-udev-load-credentials.serv  
ice  
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev-early.s  
ervice  
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-sysctl.service  
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev.service  
tmpfs           1.9G   8.0K  1.9G   1% /tmp  
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup.service  
tmpfs           1.0M   0    1.0M   0% /run/credentials/getty@tty1.service  
tmpfs           388M  120K  388M   1% /run/user/1000  
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
```

2. **uname -a:** This command displays system information like kernel version and architecture.



A terminal window titled 'himara@vbox: ~' showing the command 'uname -a' and its output. The output displays system information including the kernel version (6.8.11-amd64), architecture (x86_64), and distribution (Kali Linux).

```
(himara@vbox)-[~]  
$ uname -a  
Linux vbox 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64 GNU  
/Linux  
  
(himara@vbox)-[~]  
$
```

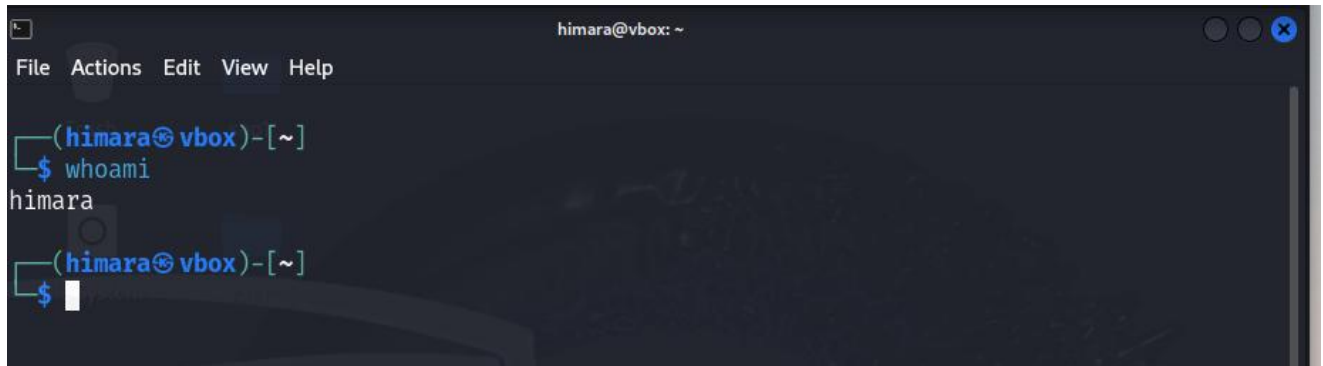
3. **free:** This command displays memory usage details of main memory (RAM) and swap memory. The free command will show information pertaining to total memory, used memory, free memory, shared memory, memory that is buffered, memory that is cached, and available memory. The default will be kibibytes and that is the same as 1024 bytes in amount.



A terminal window titled 'himara@vbox: ~' showing the command 'free' and its output. The output displays memory usage details for main memory (RAM) and swap memory. A 'Computer' status bar in the top right corner shows '78%'.

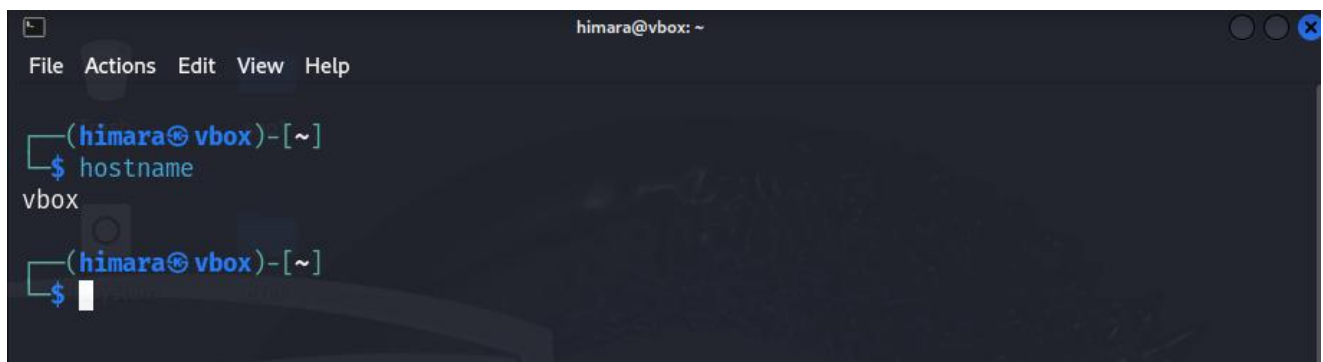
```
(himara@vbox)-[~]  
$ free  
  
              total        used        free      shared  buff/cache   available  
Mem:           3967276      904556      2795992        28648        511096      3062720  
Swap:           998396           0         998396  
  
(himara@vbox)-[~]  
$
```

4. **whoami** – **Print effective user name:** Print the user name associated with the current effective user ID.

A terminal window titled 'himara@vbox: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(himara@vbox)-[~]'. The user enters '\$ whoami' and the output 'himara' is displayed. The prompt returns to '(himara@vbox)-[~]' with a new line starting '\$' and a cursor.

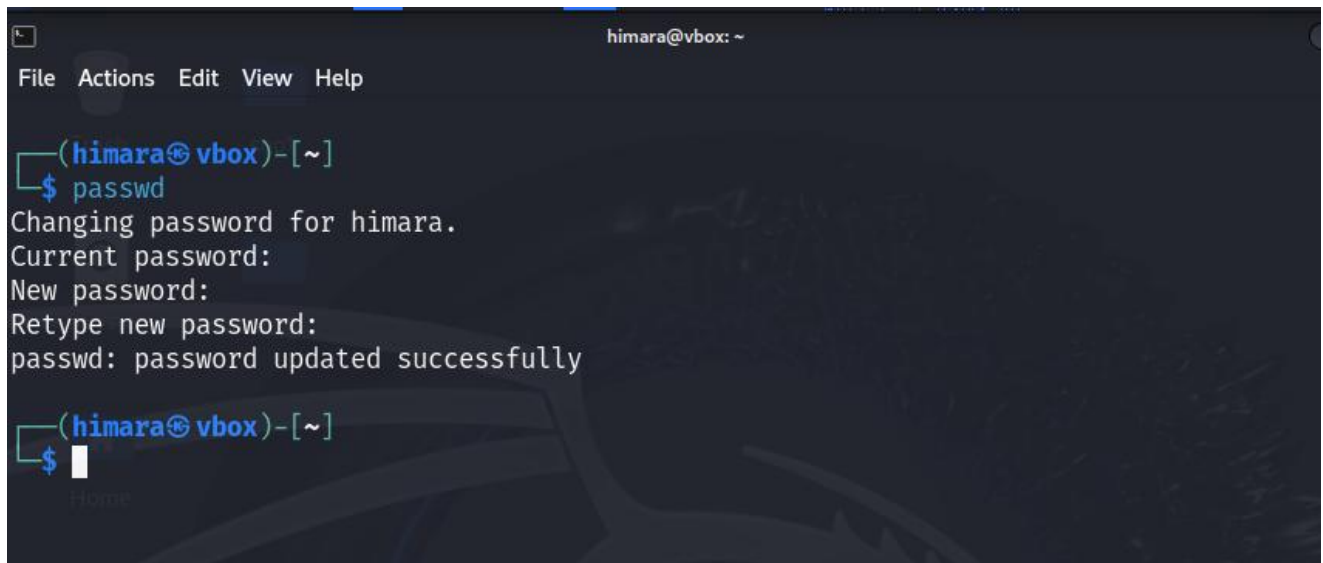
```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ whoami  
himara  
(himara@vbox)-[~]  
$
```

5. **hostname**: This command is use to display the system's DNS name, and to display or set its hostname or NIS domain name.

A terminal window titled 'himara@vbox: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(himara@vbox)-[~]'. The user enters '\$ hostname' and the output 'vbox' is displayed. The prompt returns to '(himara@vbox)-[~]' with a new line starting '\$' and a cursor.

```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ hostname  
vbox  
(himara@vbox)-[~]  
$
```

6. **passwd:** This command is use to change a user's password. Normal users can change their own and root can change any account. It asks the current password (if necessary), followed by twice the new password, and checks complexity and validity before actually changing it.



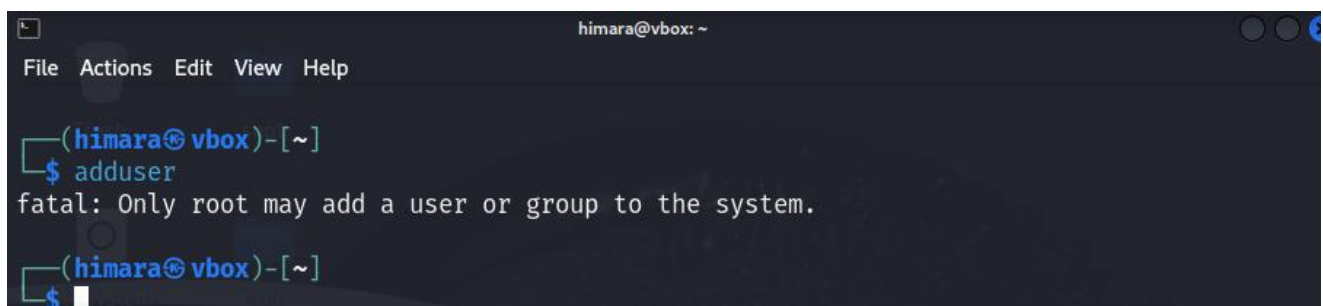
```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ passwd  
Changing password for himara.  
Current password:  
New password:  
Retype new password:  
passwd: password updated successfully  
(himara@vbox)-[~]  
$
```

7. **who:** This command shows the lists all currently logged-in users and their session details.



```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ who  
himara    tty7          2025-04-22 15:00 (:0)  
(himara@vbox)-[~]  
$
```

8. **adduser:** Adds a new user. It also creates a home directory for the user.



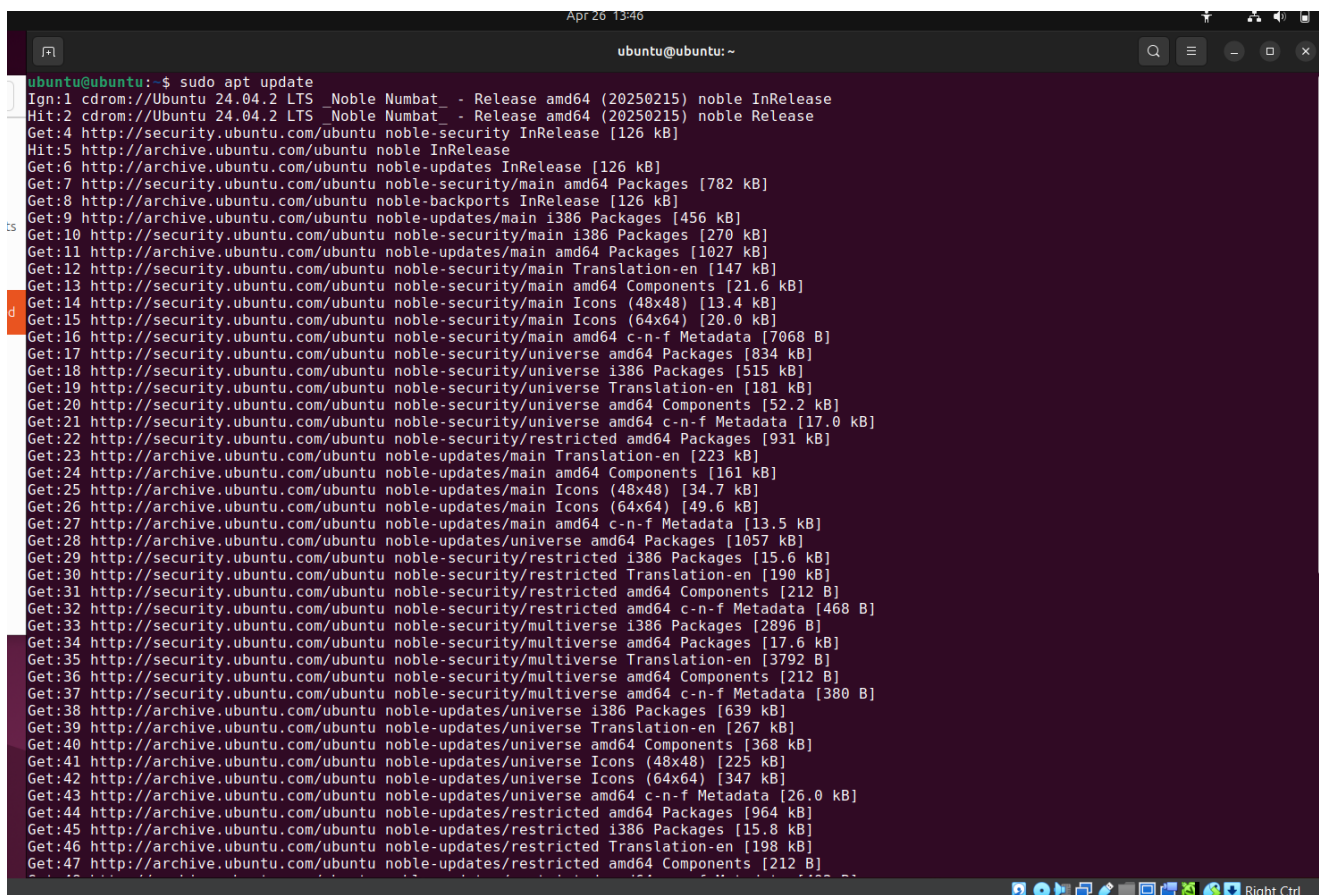
```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~]  
$ adduser  
fatal: Only root may add a user or group to the system.  
(himara@vbox)-[~]  
$
```

2. DHCP, DNS and NTP Services

1. DHCP (Dynamic Host Configuration Protocol)

1. Update the DHCP

First, execute 'apt update' to update the package index.



```
ubuntu@ubuntu: ~  
$ sudo apt update  
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease  
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release  
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease  
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [782 kB]  
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [456 kB]  
Get:10 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [270 kB]  
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1027 kB]  
Get:12 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [147 kB]  
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]  
Get:14 http://security.ubuntu.com/ubuntu noble-security/main Icons (48x48) [13.4 kB]  
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Icons (64x64) [20.0 kB]  
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [7068 B]  
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [834 kB]  
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [515 kB]  
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [181 kB]  
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]  
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]  
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [931 kB]  
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [223 kB]  
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]  
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (48x48) [34.7 kB]  
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (64x64) [49.6 kB]  
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [13.5 kB]  
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1057 kB]  
Get:29 http://security.ubuntu.com/ubuntu noble-security/restricted i386 Packages [15.6 kB]  
Get:30 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [190 kB]  
Get:31 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]  
Get:32 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [468 B]  
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse i386 Packages [2896 B]  
Get:34 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.6 kB]  
Get:35 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]  
Get:36 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]  
Get:37 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]  
Get:38 http://archive.ubuntu.com/ubuntu noble-updates/universe i386 Packages [639 kB]  
Get:39 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [267 kB]  
Get:40 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [368 kB]  
Get:41 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (48x48) [225 kB]  
Get:42 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (64x64) [347 kB]  
Get:43 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [26.0 kB]  
Get:44 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [964 kB]  
Get:45 http://archive.ubuntu.com/ubuntu noble-updates/restricted i386 Packages [15.8 kB]  
Get:46 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [198 kB]  
Get:47 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
```

```

Get:39 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [207 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [368 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (48x48) [225 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (64x64) [347 kB]
Get:43 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [26.0 kB]
Get:44 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [964 kB]
Get:45 http://archive.ubuntu.com/ubuntu noble-updates/restricted i386 Packages [15.8 kB]
Get:46 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [198 kB]
Get:47 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:48 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [492 B]
Get:49 http://archive.ubuntu.com/ubuntu noble-updates/multiverse i386 Packages [3648 B]
Get:50 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [21.5 kB]
Get:51 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [4788 B]
Get:52 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:53 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [592 B]
Get:54 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [39.1 kB]
Get:55 http://archive.ubuntu.com/ubuntu noble-backports/main i386 Packages [31.8 kB]
Get:56 http://archive.ubuntu.com/ubuntu noble-backports/main Translation-en [8676 B]
Get:57 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7080 B]
Get:58 http://archive.ubuntu.com/ubuntu noble-backports/main Icons (48x48) [9518 B]
Get:59 http://archive.ubuntu.com/ubuntu noble-backports/main Icons (64x64) [11.2 kB]
Get:60 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [272 B]
Get:61 http://archive.ubuntu.com/ubuntu noble-backports/universe i386 Packages [14.0 kB]
Get:62 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [27.1 kB]
Get:63 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [16.5 kB]
Get:64 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [16.3 kB]
Get:65 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (48x48) [20.4 kB]
Get:66 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (64x64) [28.7 kB]
Get:67 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1304 B]
Get:68 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:69 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 10.7 MB in 15s (698 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
250 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$

```

2.Install the DHCP server

Start by obtaining the **isc-dhcp-server** package. After that, execute the command '**apt install isc-dhcp-server**' to install and set it up on system.

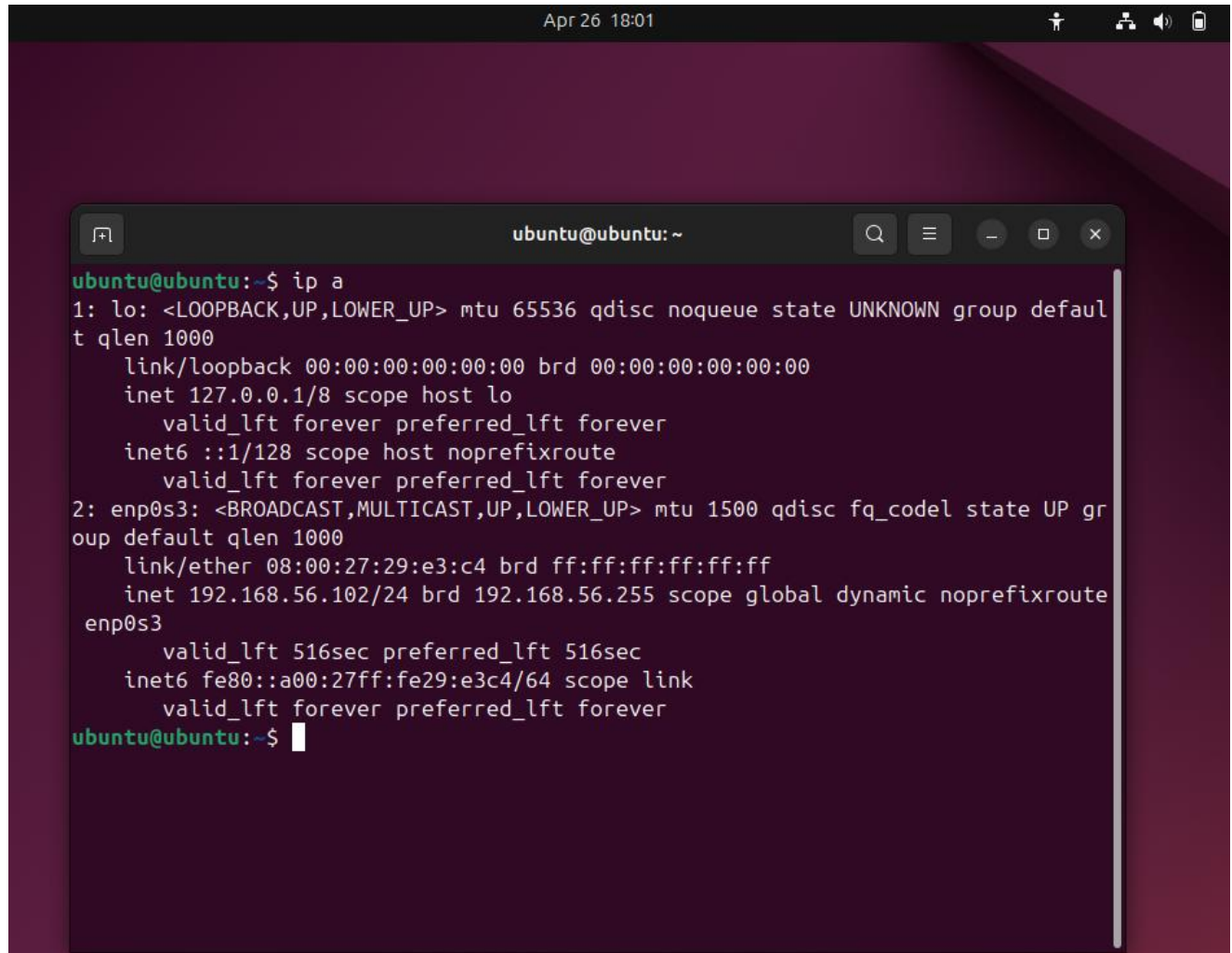
```

Apr 26 13:49
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo apt install isc-dhcp-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  isc-dhcp-common
Suggested packages:
  isc-dhcp-server-ldap policycoreutils
The following NEW packages will be installed:
  isc-dhcp-common isc-dhcp-server
0 upgraded, 2 newly installed, 0 to remove and 250 not upgraded.
Need to get 1281 kB of archives.
After this operation, 4281 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-server amd64 4.4.3-P1-4ubuntu2 [1236 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-common amd64 4.4.3-P1-4ubuntu2 [45.8 kB]
Fetched 1281 kB in 3s (401 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database ... 212032 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Selecting previously unselected package isc-dhcp-common.
Preparing to unpack .../isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Setting up isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Generating /etc/default/isc-dhcp-server...
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server.service → /usr/lib/systemd/system/isc-dhcp-server.service.
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.service → /usr/lib/systemd/system/isc-dhcp-server6.service.
Setting up isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu@ubuntu:~$

```


3. Configure the DHCP server

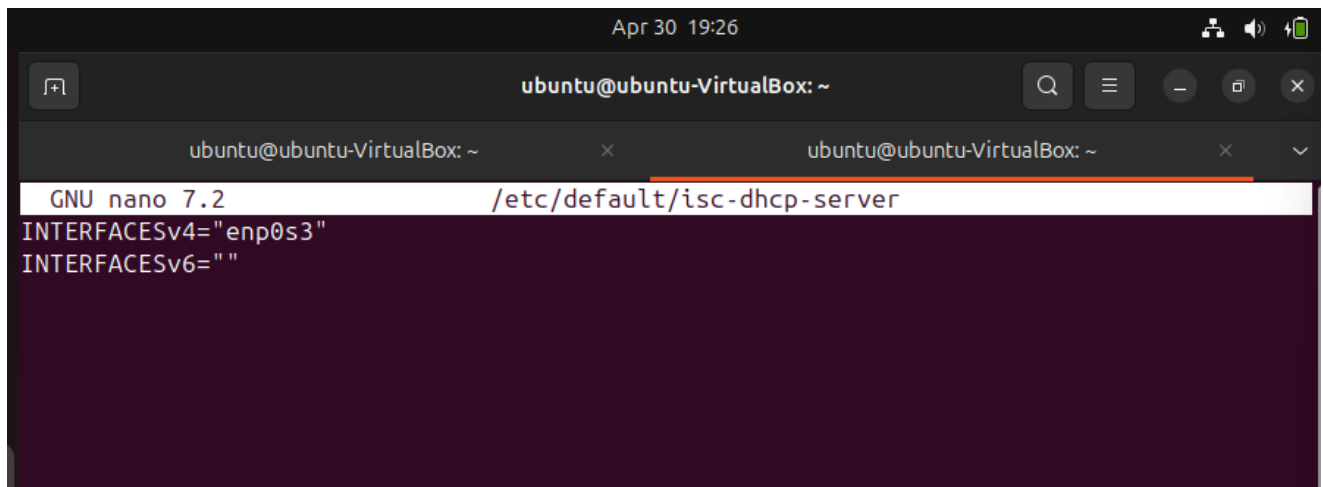
I need to get the IP address and network interface using the **'ip a'** command.



The image shows a terminal window on an Ubuntu system. The window title is 'ubuntu@ubuntu: ~'. The terminal displays the output of the command 'ip a'. The output shows details for two network interfaces: 'lo' (loopback) and 'enp0s3' (Ethernet). The 'lo' interface has an IP address of 127.0.0.1. The 'enp0s3' interface has an IP address of 192.168.56.102. The terminal text is as follows:

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:29:e3:c4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
        valid_lft 516sec preferred_lft 516sec
    inet6 fe80::a00:27ff:fe29:e3c4/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

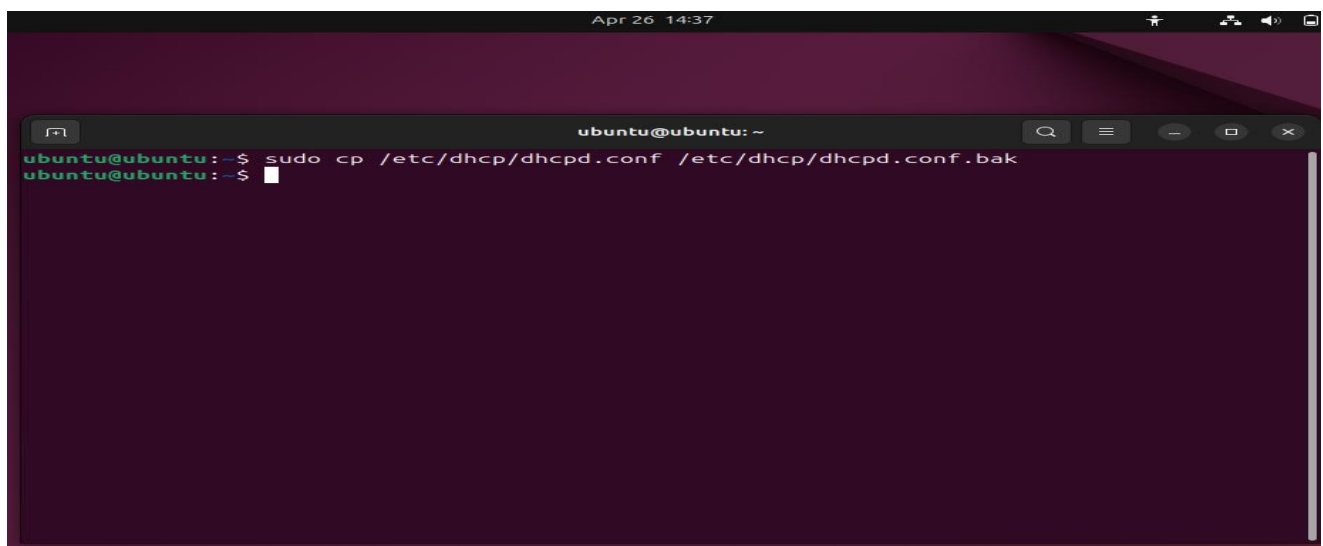
Edit the */etc/default/isc-dhcp-server* file to specify which interface the DHCP server should listen on and set the interface to “enp0s3”.



The screenshot shows a terminal window titled 'ubuntu@ubuntu-VirtualBox: ~' with a timestamp of 'Apr 30 19:26'. The terminal is running the nano editor to edit the file */etc/default/isc-dhcp-server*. The editor's status bar at the top indicates 'GNU nano 7.2' and the file path. The content of the file is as follows:

```
INTERFACESv4="enp0s3"
INTERFACESv6=""
```

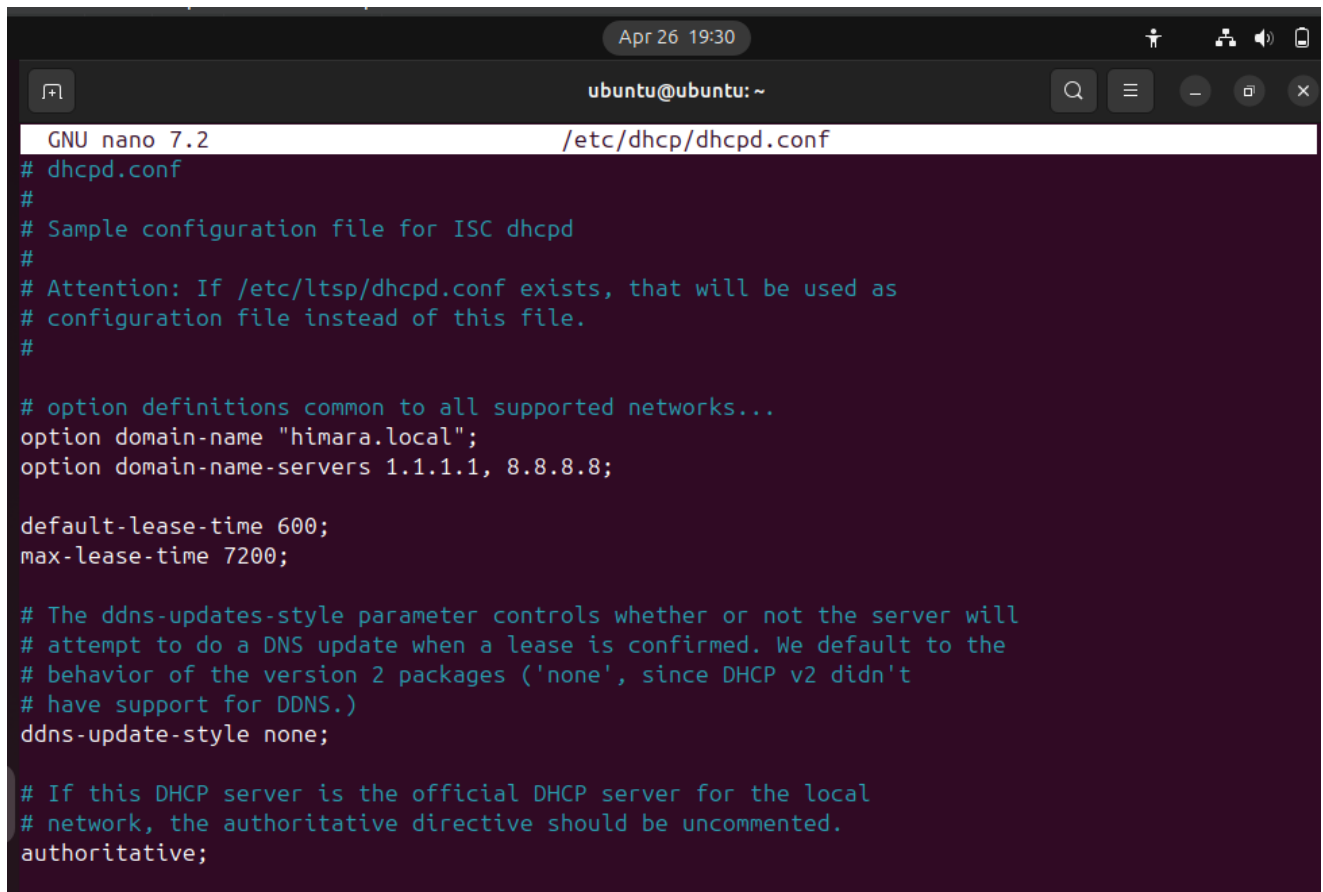
After installing the DHCP server the next step is to configure the DHCP server. - The configuration file for ISC DHCP is located at */etc/dhcp/dhcpd.conf*. Before making any changes, it's wise to create a backup of the original file using cp.



The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~' with a timestamp of 'Apr 26 14:37'. The terminal displays the command to create a backup of the DHCP configuration file:

```
ubuntu@ubuntu:~$ sudo cp /etc/dhcp/dhcpd.conf /etc/dhcp/dhcpd.conf.bak
ubuntu@ubuntu:~$
```


Next edit the `/etc/dhcp/dhcpd.conf` file to add or modify the configuration according to your network. Make the following modifications, then save the file and exit (ctrl+O press Enter and ctrl+X).

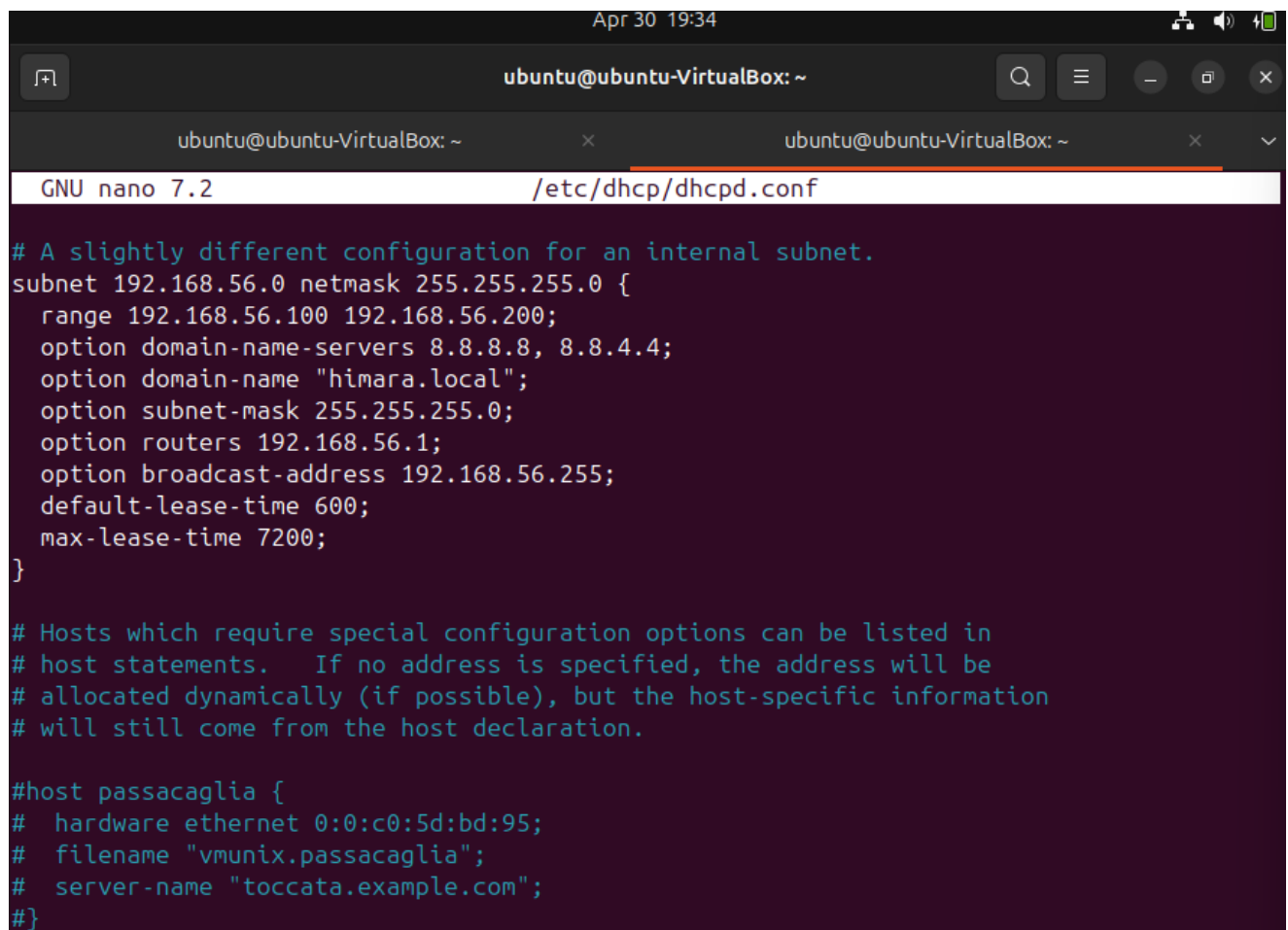


```
Apr 26 19:30
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/dhcp/dhcpd.conf
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#
# Attention: If /etc/ltsp/dhcpd.conf exists, that will be used as
# configuration file instead of this file.
#
# option definitions common to all supported networks...
option domain-name "himara.local";
option domain-name-servers 1.1.1.1, 8.8.8.8;

default-lease-time 600;
max-lease-time 7200;

# The ddns-updates-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;
```



The screenshot shows a terminal window titled 'ubuntu@ubuntu-VirtualBox: ~' with a date and time of 'Apr 30 19:34'. The terminal is running the 'nano' text editor to edit the file '/etc/dhcp/dhcpd.conf'. The configuration file content is as follows:

```
GNU nano 7.2 /etc/dhcp/dhcpd.conf

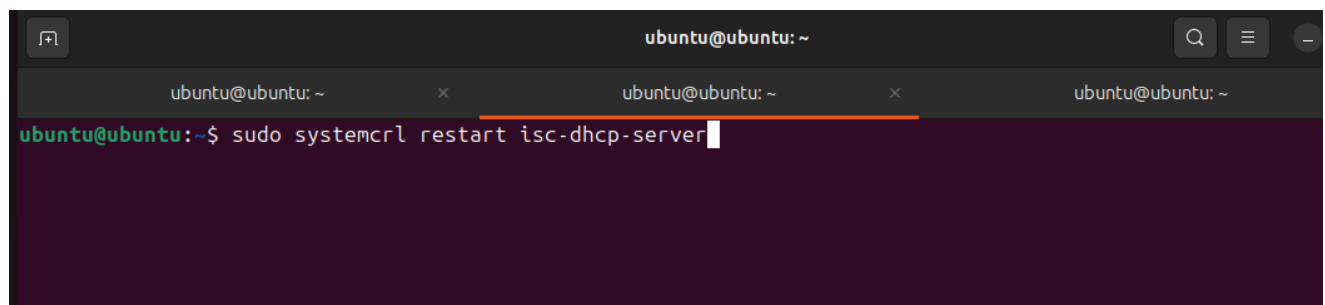
# A slightly different configuration for an internal subnet.
subnet 192.168.56.0 netmask 255.255.255.0 {
    range 192.168.56.100 192.168.56.200;
    option domain-name-servers 8.8.8.8, 8.8.4.4;
    option domain-name "himara.local";
    option subnet-mask 255.255.255.0;
    option routers 192.168.56.1;
    option broadcast-address 192.168.56.255;
    default-lease-time 600;
    max-lease-time 7200;
}

# Hosts which require special configuration options can be listed in
# host statements.  If no address is specified, the address will be
# allocated dynamically (if possible), but the host-specific information
# will still come from the host declaration.

#host passacaglia {
#   hardware ethernet 0:0:c0:5d:bd:95;
#   filename "vmunix.passacaglia";
#   server-name "toccata.example.com";
#}
```

4. Restart the DHCP server

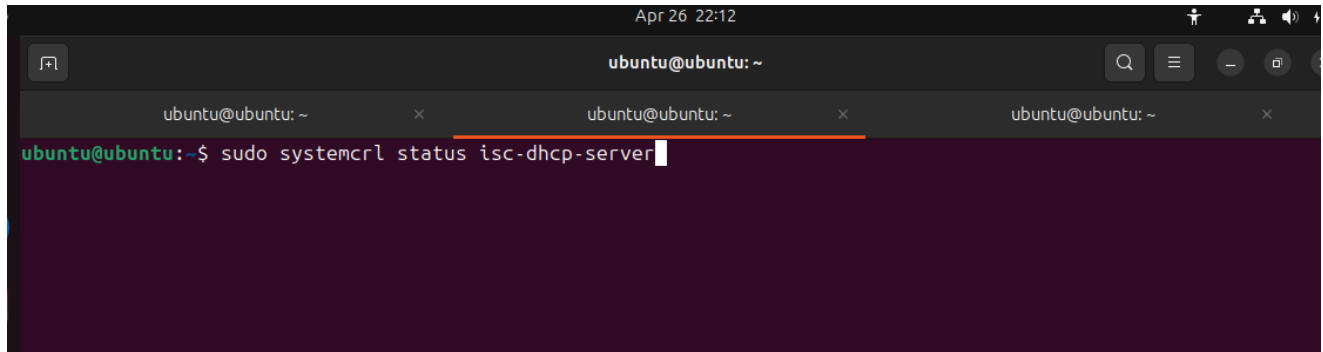
After finish the necessary configuration restart the DHCP server and save the changes.



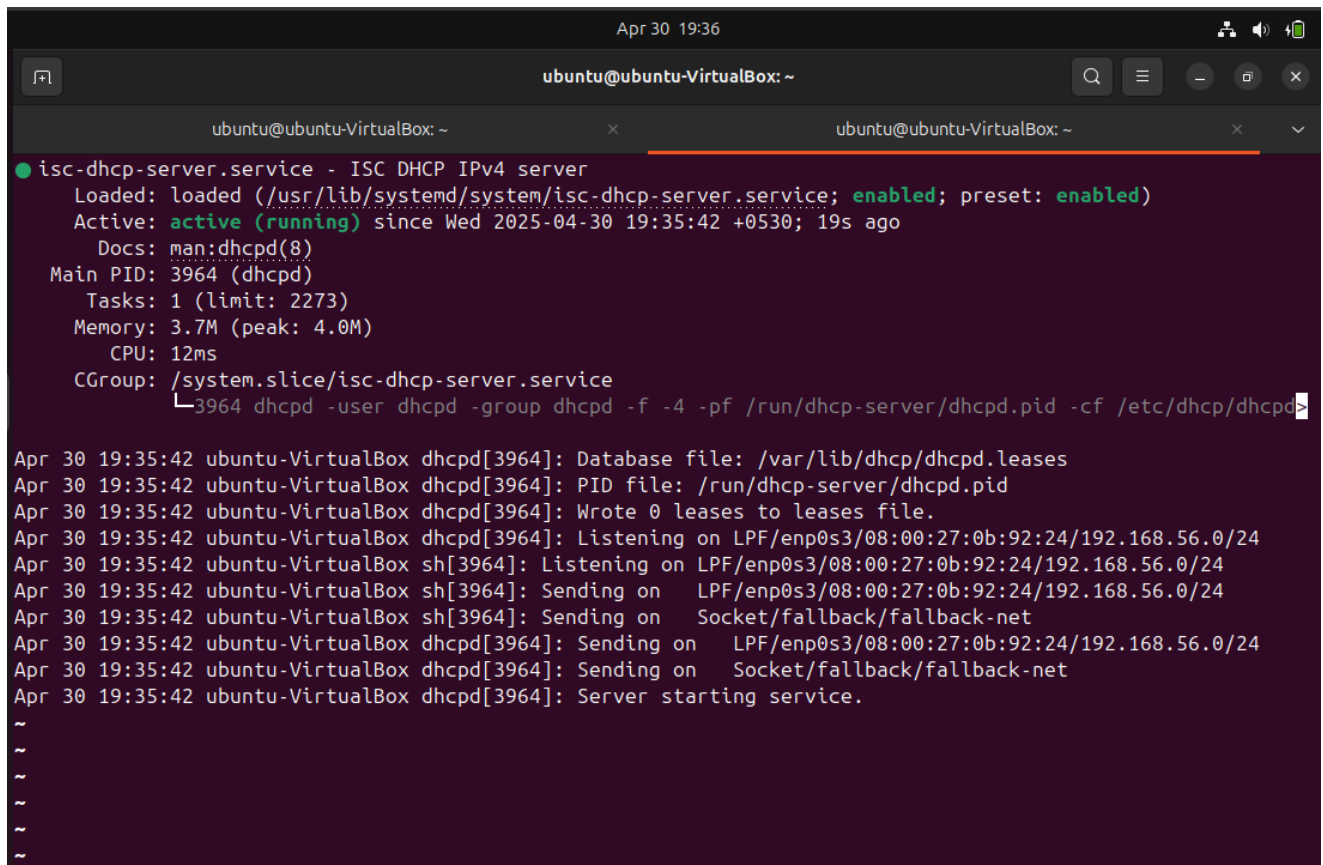
The screenshot shows a terminal window titled 'ubuntu@ubuntu: ~'. The terminal is running the command 'sudo systemctl restart isc-dhcp-server' to restart the DHCP service. The command prompt is 'ubuntu@ubuntu:~\$' and the command is 'sudo systemctl restart isc-dhcp-server'.

5. Check the Status of the Service

The service ought to appear as active (running) on your list. Examine the configuration and log files for problems if there are any failures.



```
Apr 26 22:12
ubuntu@ubuntu: ~
ubuntu@ubuntu: ~
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo systemctl status isc-dhcp-server
```



```
Apr 30 19:36
ubuntu@ubuntu-VirtualBox: ~
ubuntu@ubuntu-VirtualBox: ~
ubuntu@ubuntu-VirtualBox: ~
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-04-30 19:35:42 +0530; 19s ago
     Docs: man:dhcpd(8)
    Main PID: 3964 (dhcpd)
      Tasks: 1 (limit: 2273)
    Memory: 3.7M (peak: 4.0M)
       CPU: 12ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─3964 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf

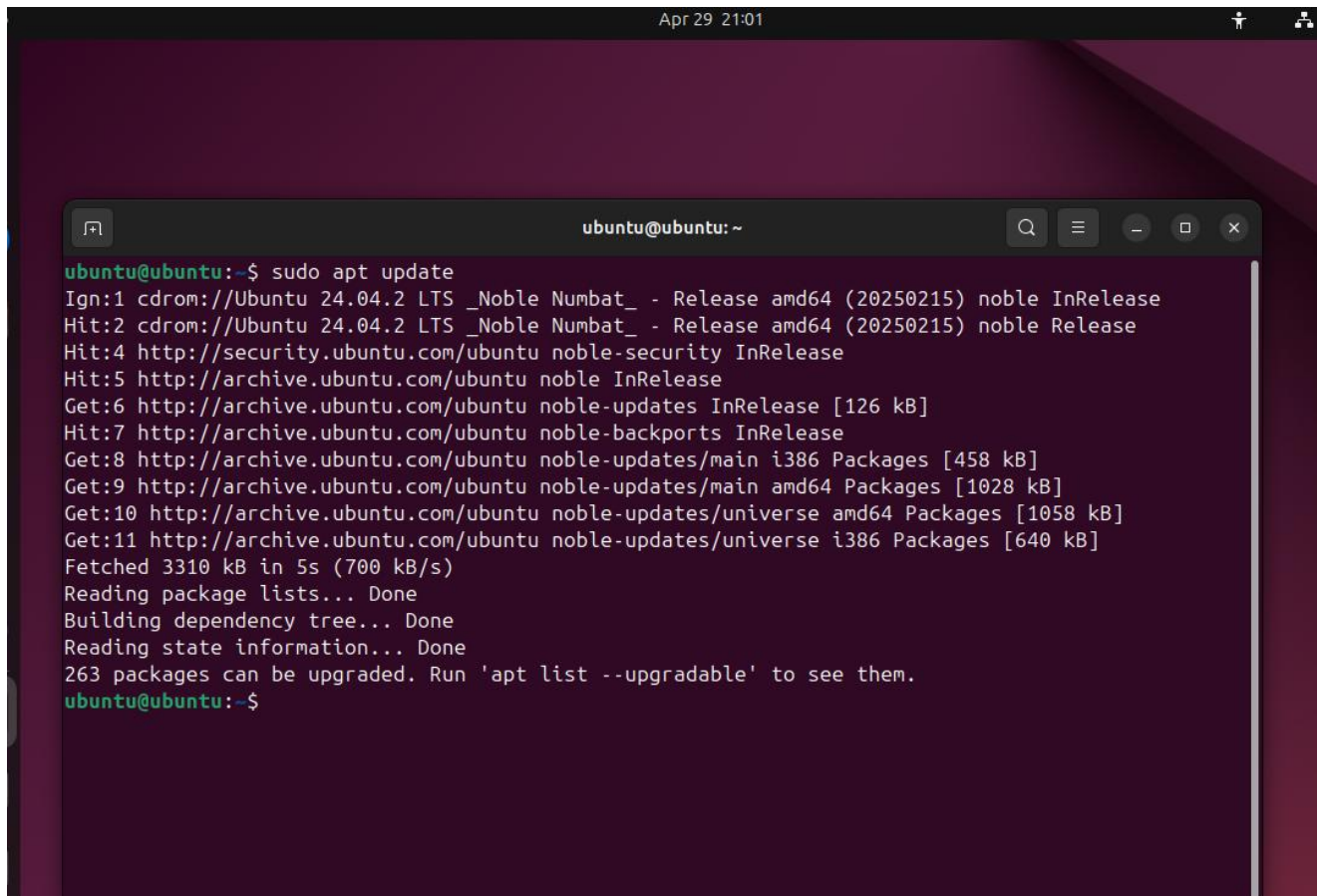
Apr 30 19:35:42 ubuntu-VirtualBox dhcpd[3964]: Database file: /var/lib/dhcp/dhcpd.leases
Apr 30 19:35:42 ubuntu-VirtualBox dhcpd[3964]: PID file: /run/dhcp-server/dhcpd.pid
Apr 30 19:35:42 ubuntu-VirtualBox dhcpd[3964]: Wrote 0 leases to leases file.
Apr 30 19:35:42 ubuntu-VirtualBox dhcpd[3964]: Listening on LPF/enp0s3/08:00:27:0b:92:24/192.168.56.0/24
Apr 30 19:35:42 ubuntu-VirtualBox sh[3964]: Listening on LPF/enp0s3/08:00:27:0b:92:24/192.168.56.0/24
Apr 30 19:35:42 ubuntu-VirtualBox sh[3964]: Sending on LPF/enp0s3/08:00:27:0b:92:24/192.168.56.0/24
Apr 30 19:35:42 ubuntu-VirtualBox sh[3964]: Sending on Socket/fallback/fallback-net
Apr 30 19:35:42 ubuntu-VirtualBox dhcpd[3964]: Sending on LPF/enp0s3/08:00:27:0b:92:24/192.168.56.0/24
Apr 30 19:35:42 ubuntu-VirtualBox dhcpd[3964]: Sending on Socket/fallback/fallback-net
Apr 30 19:35:42 ubuntu-VirtualBox dhcpd[3964]: Server starting service.
~
~
~
~
~
```

2. DNS (Domain Name Server)

Configuration of DNS server

1. Update the Ubuntu system

This will update the system and refreshes your system's package list from the Ubuntu repositories.

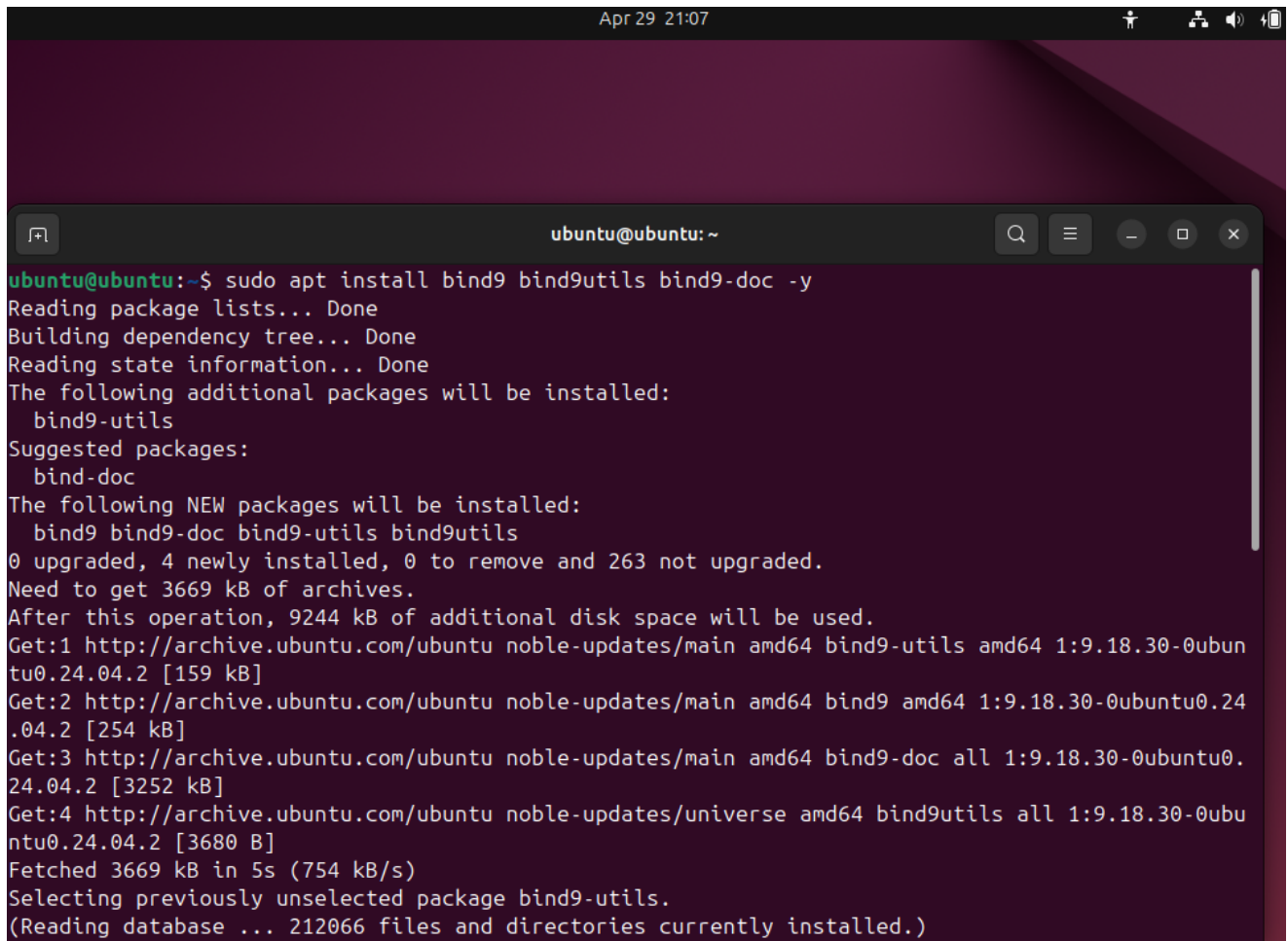
A terminal window titled 'ubuntu@ubuntu: ~' is shown against a dark purple background. The window displays the output of the 'sudo apt update' command. The output shows the system fetching package lists from various Ubuntu repositories, including the main archive, security updates, and universe packages. It reports that 263 packages can be upgraded. The terminal text is as follows:

```
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [458 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1028 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1058 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe i386 Packages [640 kB]
Fetched 3310 kB in 5s (700 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
263 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$
```

2. Install the BIND9

Now, get the package and run the command ‘sudo apt install bind9 bind9utils bind9-doc -y’ to set it up.

- **Bind (Berkeley Internet Name Domain)** is a widely used and highly reliable DNS server software.
- **bind9**: The DNS server software.
- **bind9utils**: Useful command-line utilities for BIND9.
- **bind9-doc**: Documentation for BIND9 (optional).



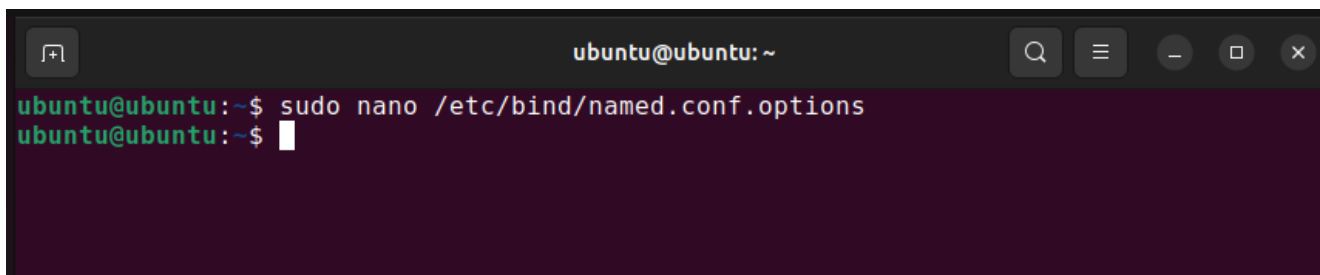
```
Apr 29 21:07
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo apt install bind9 bind9utils bind9-doc -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bind9-utils
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9 bind9-doc bind9-utils bind9utils
0 upgraded, 4 newly installed, 0 to remove and 263 not upgraded.
Need to get 3669 kB of archives.
After this operation, 9244 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-utils amd64 1:9.18.30-0ubuntu0.24.04.2 [159 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9 amd64 1:9.18.30-0ubuntu0.24.04.2 [254 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-doc all 1:9.18.30-0ubuntu0.24.04.2 [3252 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 bind9utils all 1:9.18.30-0ubuntu0.24.04.2 [3680 B]
Fetched 3669 kB in 5s (754 kB/s)
Selecting previously unselected package bind9-utils.
(Reading database ... 212066 files and directories currently installed.)
```

```
.2 [254 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-doc all 1:9.18.30-0ubuntu0.24.04.2 [3252 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 bind9utils all 1:9.18.30-0ubuntu0.24.04.2 [3680 B]
Fetched 3669 kB in 5s (754 kB/s)
Selecting previously unselected package bind9-utils.
(Reading database ... 212066 files and directories currently installed.)
Preparing to unpack .../bind9-utils_1%3a9.18.30-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9-utils (1:9.18.30-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1%3a9.18.30-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9 (1:9.18.30-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9-doc.
Preparing to unpack .../bind9-doc_1%3a9.18.30-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9-doc (1:9.18.30-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9utils.
Preparing to unpack .../bind9utils_1%3a9.18.30-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9utils (1:9.18.30-0ubuntu0.24.04.2) ...
Setting up bind9-doc (1:9.18.30-0ubuntu0.24.04.2) ...
Setting up bind9-utils (1:9.18.30-0ubuntu0.24.04.2) ...
Setting up bind9 (1:9.18.30-0ubuntu0.24.04.2) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `bind' (GID 126) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user `bind' (UID 123) ...
info: Adding new user `bind' (UID 123) with group `bind' ...
info: Not creating home directory `/var/cache/bind'.
wrote key file "/etc/bind/rndc.key"
named-resolvconf.service is a disabled or a static unit, not starting it.
Created symlink /etc/systemd/system/bind9.service → /usr/lib/systemd/system/named.service.
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
Setting up bind9utils (1:9.18.30-0ubuntu0.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
ubuntu@ubuntu:~$
```

3. Configure BIND9

- The named.conf file is BIND 9's primary configuration file.
- We can declare options we need for our configuration.

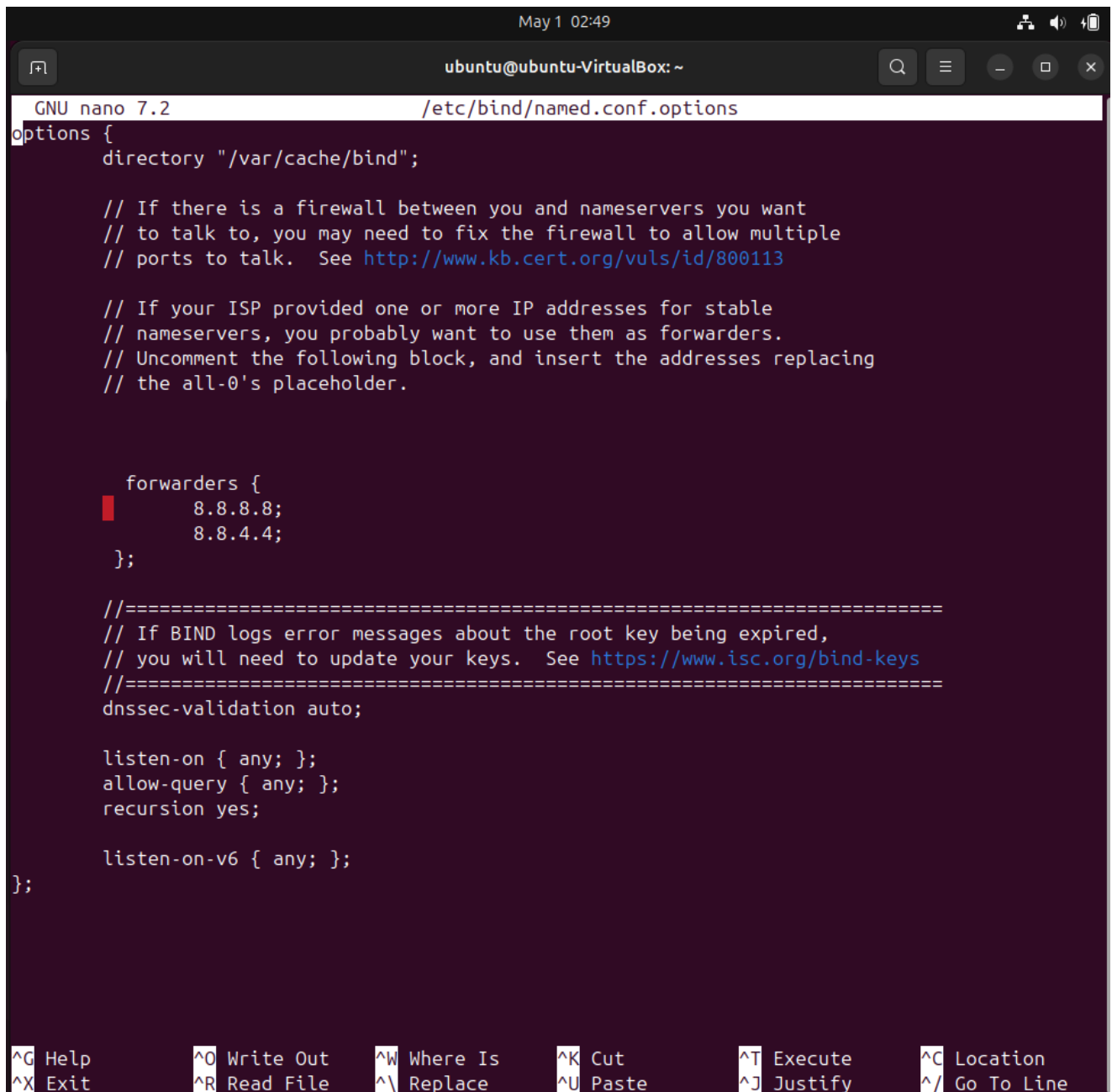


```
ubuntu@ubuntu:~$ sudo nano /etc/bind/named.conf.options
ubuntu@ubuntu:~$
```

- After that open the file and do the following editors and the necessary codes should be uncommented and be modified.

8.8.8.8: Google DNS

8.8.4.4: Google DNS



The screenshot shows a terminal window titled 'ubuntu@ubuntu-VirtualBox: ~' with a timestamp of 'May 1 02:49'. The terminal is running the GNU nano 7.2 editor, editing the file '/etc/bind/named.conf.options'. The content of the file is as follows:

```
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //=====
    dnssec-validation auto;

    listen-on { any; };
    allow-query { any; };
    recursion yes;

    listen-on-v6 { any; };
};
```

At the bottom of the terminal, there is a status bar with various keyboard shortcuts: ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^_ Go To Line.

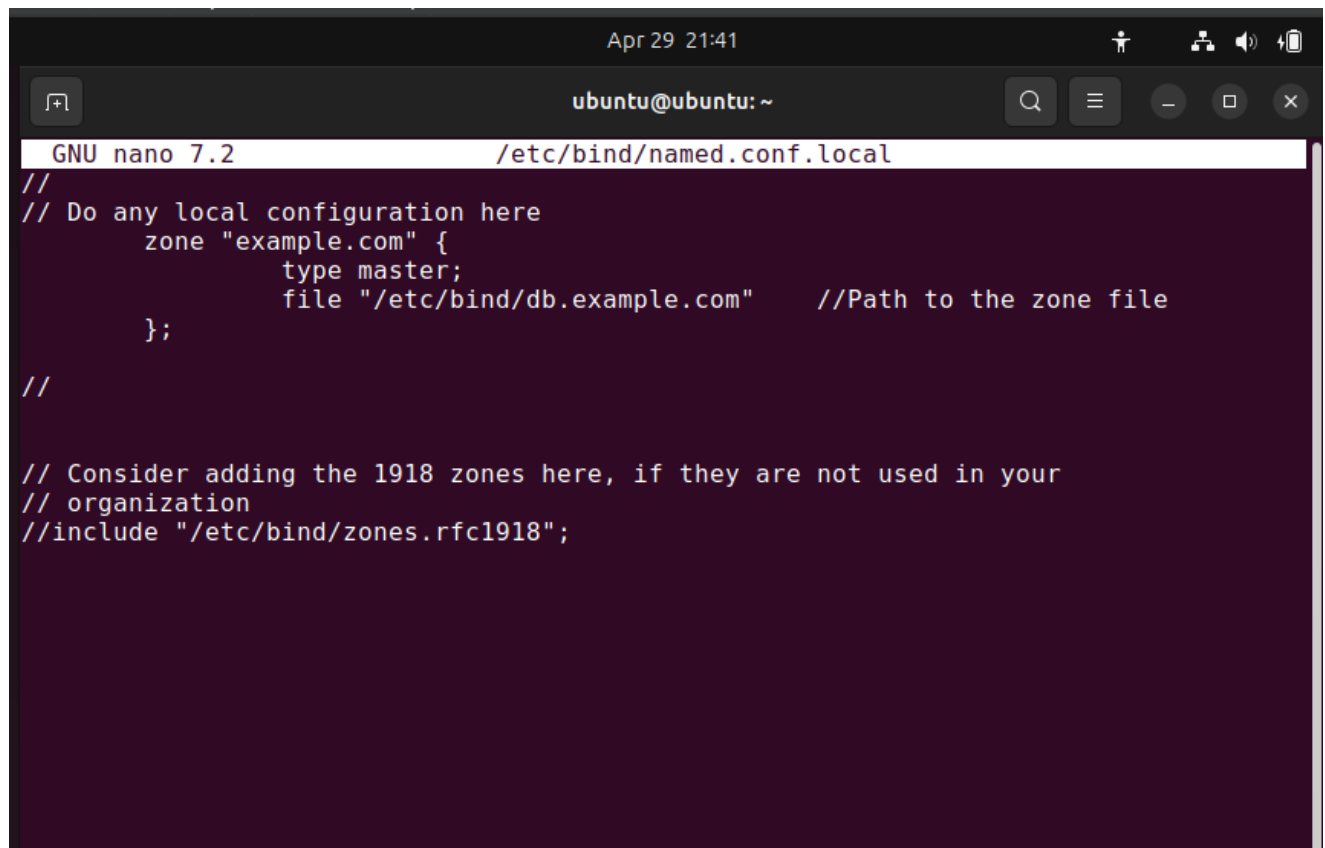
These are changed so that they send DNS requests that your server cannot resolve to other DNS servers.

4. Configure Zone Files

- To define your own DNS zones (for internal or external domains), you'll need to add the zone definition in the *named.conf.local* file.
- A **DNS zone** represents a portion of the DNS namespace and contains resource records that map domain names to IP addresses.

```
ubuntu@ubuntu-VirtualBox:~$ sudo nano /etc/bind/named.conf.local
```

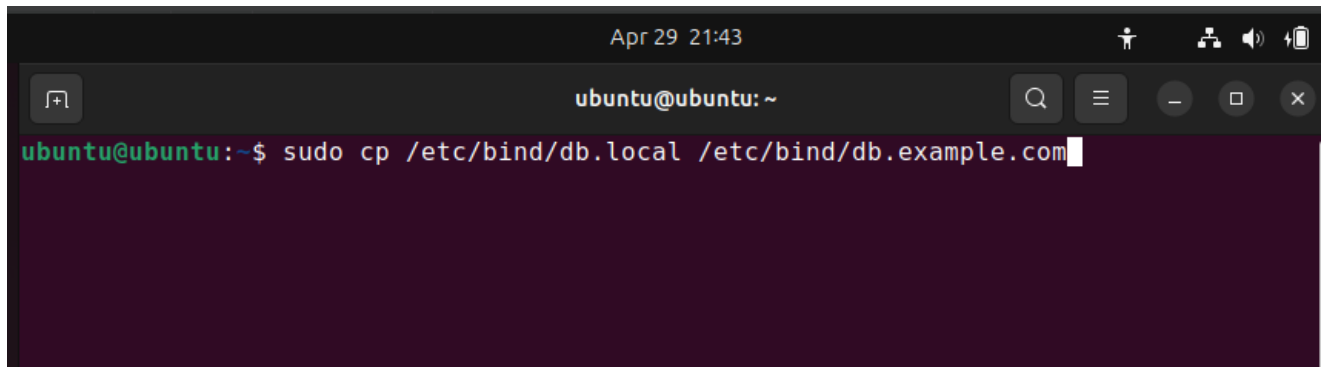
Edit the file following instructions to define the zone.



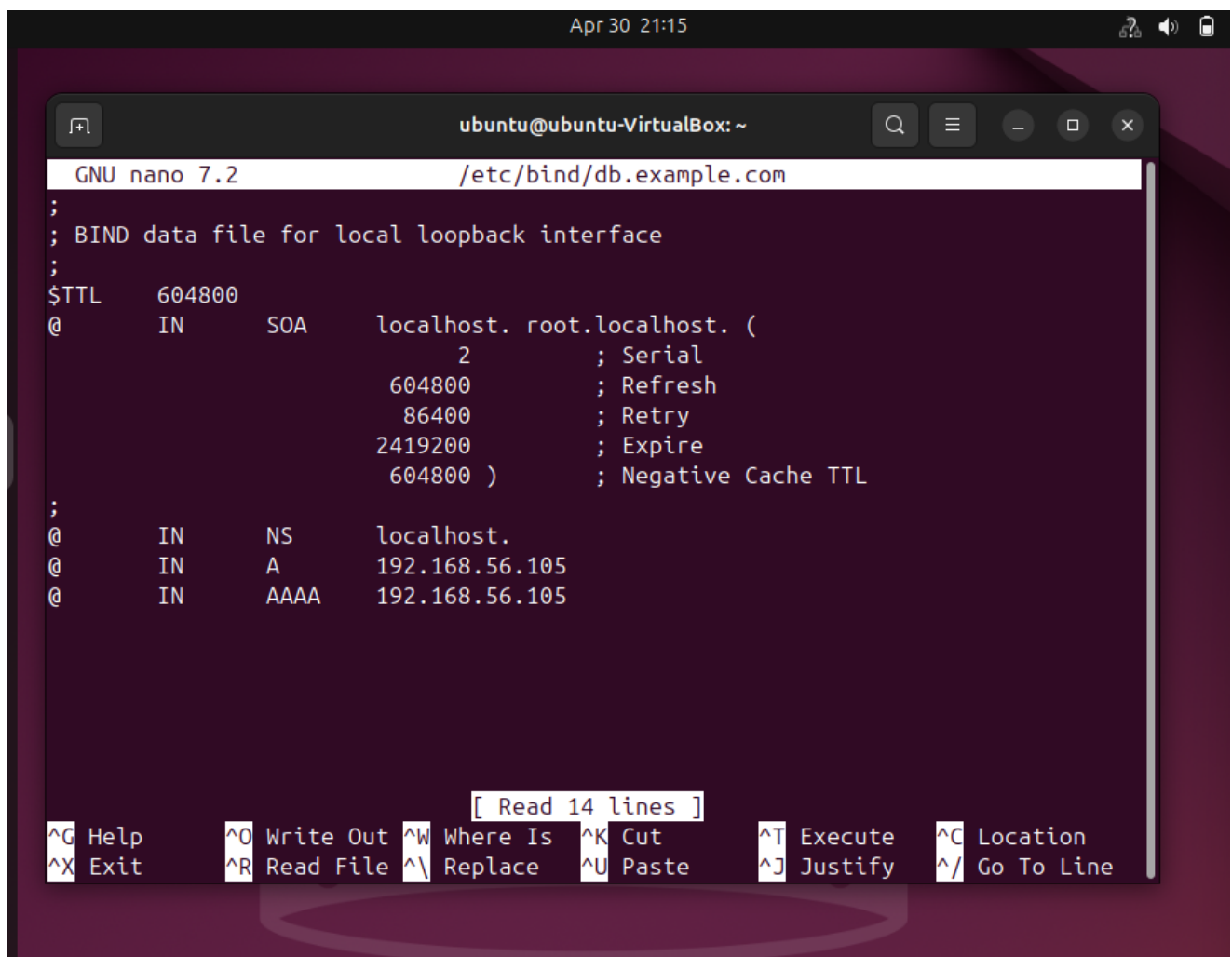
```
Apr 29 21:41
ubuntu@ubuntu: ~
GNU nano 7.2 /etc/bind/named.conf.local
//
// Do any local configuration here
    zone "example.com" {
        type master;
        file "/etc/bind/db.example.com" //Path to the zone file
    };
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";
```


Next create a new zone file for your domain and open the newly created file and configure your DNS records.



```
Apr 29 21:43
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo cp /etc/bind/db.local /etc/bind/db.example.com
```



```
Apr 30 21:15
ubuntu@ubuntu-VirtualBox: ~
GNU nano 7.2 /etc/bind/db.example.com
;
; BIND data file for local loopback interface
;
$TTL      604800
@         IN      SOA      localhost. root.localhost. (
                        2      ; Serial
                        604800  ; Refresh
                        86400   ; Retry
                        2419200 ; Expire
                        604800  ) ; Negative Cache TTL
;
@         IN      NS       localhost.
@         IN      A        192.168.56.105
@         IN      AAAA     192.168.56.105

[ Read 14 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

The ip address is should be changed to my IP address 192.168.56.105.

5. Check the configuration

Check the syntax of BIND configuration and run the following command to check for errors in the configuration.

- “**sudo named-checkconf**” this command is used to check the zone file.
- “**sudo named-checkzone example.com /etc/bind/db.example.com**” this command is used to and verify that zone file are correctly formed.

```
ubuntu@ubuntu-VirtualBox:~$ sudo named-checkconf
ubuntu@ubuntu-VirtualBox:~$ sudo named-checkzone example.com /etc/bind/db.example.com
zone example.com/IN: loaded serial 2
OK
ubuntu@ubuntu-VirtualBox:~$
```

6. Start BIND

To apply the configuration changes, restart the DNS server.

```
ubuntu@ubuntu-VirtualBox:~$ sudo systemctl restart bind9
```

7. Enable BIND9 to start on boot

```
ubuntu@ubuntu-VirtualBox:~$ sudo systemctl enable bind9
```

Now use the '**sudo systemctl status bind9**' command and check the status of the server. OK It is (Active) running.

[illegible]

8. Check BIND9 allow for the firewall - First

We have to configure the **bind9** service to be allowed through the firewall.

```
ubuntu@ubuntu-VirtualBox:~$ sudo ufw allow 53
Rules updated
Rules updated (v6)
ubuntu@ubuntu-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu-VirtualBox:~$ sudo ufw status
Status: active

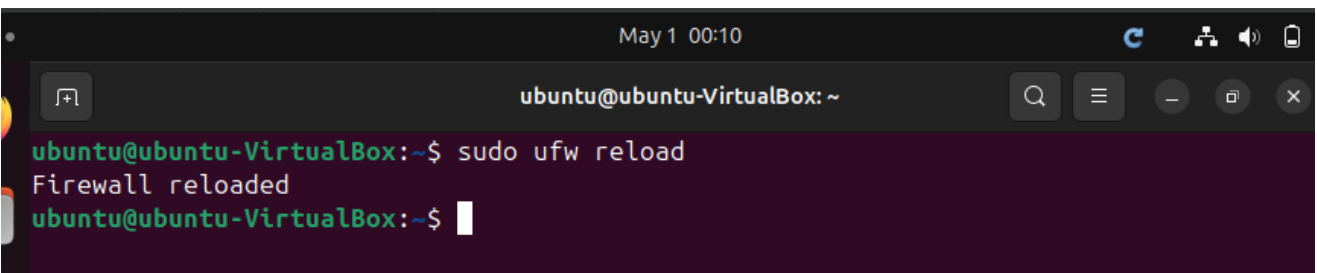
To                                Action    From
--                                -
53                                ALLOW    Anywhere
53 (v6)                          ALLOW    Anywhere (v6)

ubuntu@ubuntu-VirtualBox:~$
```

Now give a rule to BIND9 to go through the firewall.

```
ubuntu@ubuntu-VirtualBox:~$ sudo ufw allow bind9
Rule added
Rule added (v6)
ubuntu@ubuntu-VirtualBox:~$
```

Now reload the ufw.

A terminal window titled 'ubuntu@ubuntu-VirtualBox: ~' with a search bar and window controls. It shows the command 'sudo ufw reload' being executed, resulting in 'Firewall reloaded'.

```
ubuntu@ubuntu-VirtualBox:~$ sudo ufw reload
Firewall reloaded
ubuntu@ubuntu-VirtualBox:~$
```

```
ubuntu@ubuntu-VirtualBox:~$ sudo ufw status
Status: active

To Action From
--
53 ALLOW Anywhere
Bind9 ALLOW Anywhere
53 (v6) ALLOW Anywhere (v6)
Bind9 (v6) ALLOW Anywhere (v6)

ubuntu@ubuntu-VirtualBox:~$
```

9. Checking whether the DNS server works

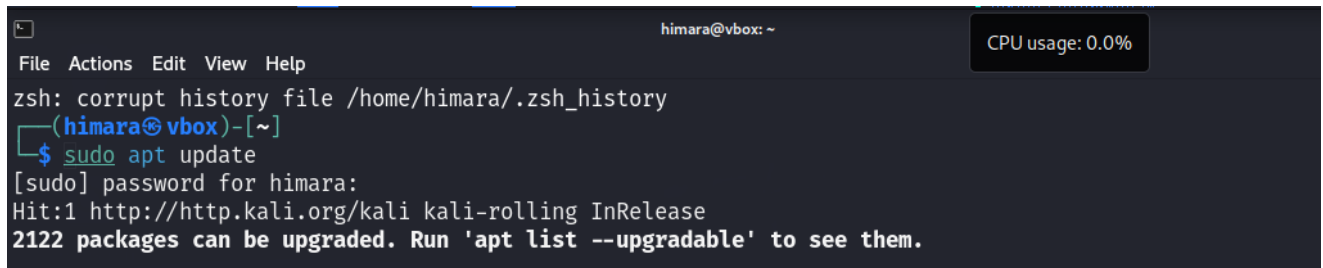
This output shows the result of a DNS query for domain `www.example.com` that has been successfully resolved by the DNS server with IP address `192.168.56.105` and which has replied with IP address `192.168.56.25`. The DNS query and response process was carried out without any errors

```
himara@vbox: ~  
File Actions Edit View Help  
zsh: corrupt history file /home/himara/.zsh_history  
(himara@vbox)-[~]  
$ dig @192.168.56.105 www.example.com  
From 192.168.56.103 icmp_seq=582 Destination Host Unreachable  
; <<>> DiG 9.20.7-1-Debian <<>> @192.168.56.105 www.example.com  
; (1 server found) icmp_seq=583 Destination Host Unreachable  
;; global options: +cmd icmp_seq=584 Destination Host Unreachable  
;; Got answer: icmp_seq=585 Destination Host Unreachable  
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 29148 icmp_seq=586 Destination Host Unreachable  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1 icmp_seq=587 Destination Host Unreachable  
From 192.168.56.103 icmp_seq=588 Destination Host Unreachable  
;; OPT PSEUDOSECTION: icmp_seq=589 Destination Host Unreachable  
; EDNS: version: 0, flags:: udp: 1232 icmp_seq=590 Destination Host Unreachable  
; COOKIE: c1afde5c29cd9a87010000006812875e000ae8929a6ef96f (good) icmp_seq=591 Destination Host Unreachable  
;; QUESTION SECTION: icmp_seq=592 Destination Host Unreachable  
;www.example.com. 3 IN stdn A on Host Unreachable  
From 192.168.56.103 icmp_seq=593 Destination Host Unreachable  
;; ANSWER SECTION: icmp_seq=594 Destination Host Unreachable  
www.example.com. 03 IN 604800 IN stdn A on Ho 192.168.56.25  
From 192.168.56.103 icmp_seq=595 Destination Host Unreachable  
;; Query time: 8 msec icmp_seq=596 Destination Host Unreachable  
;; SERVER: 192.168.56.105#53(192.168.56.105) (UDP) unreachable  
;; WHEN: Wed Apr 30 16:26:05 EDT 2025 icmp_seq=597 Destination Host Unreachable  
;; MSG SIZE rcvd: 88 icmp_seq=598 Destination Host Unreachable  
From 192.168.56.103 icmp_seq=599 Destination Host Unreachable  
From 192.168.56.103 icmp_seq=600 Destination Host Unreachable  
(himara@vbox)-[~]  
$
```

3. NTP (Network Time Protocol)

1. Update the Linux system

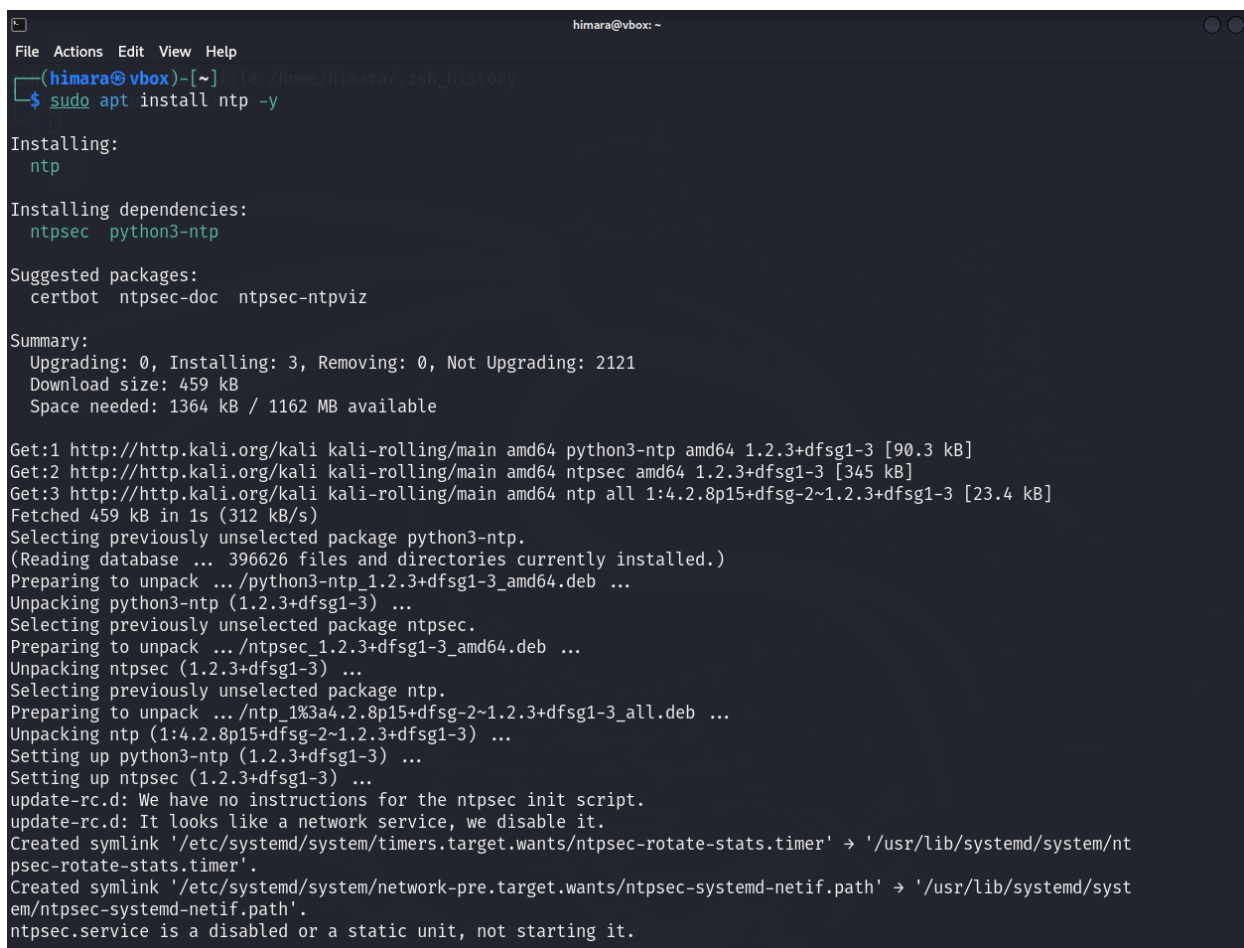
“**sudo apt update**” using this command you can update the your Linux system.



```
himara@vbox: ~
File Actions Edit View Help
zsh: corrupt history file /home/himara/.zsh_history
(himara@vbox)-[~]
$ sudo apt update
[sudo] password for himara:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
2122 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

2. Install NTP

“**sudo apt install ntp -y**” using this command you can install the NTP package. The NTP package provides both the NTP daemon and the tools necessary to synchronize time.



```
himara@vbox: ~
File Actions Edit View Help
(himara@vbox)-[~]
$ sudo apt install ntp -y

Installing:
ntp

Installing dependencies:
ntpsec python3-ntp

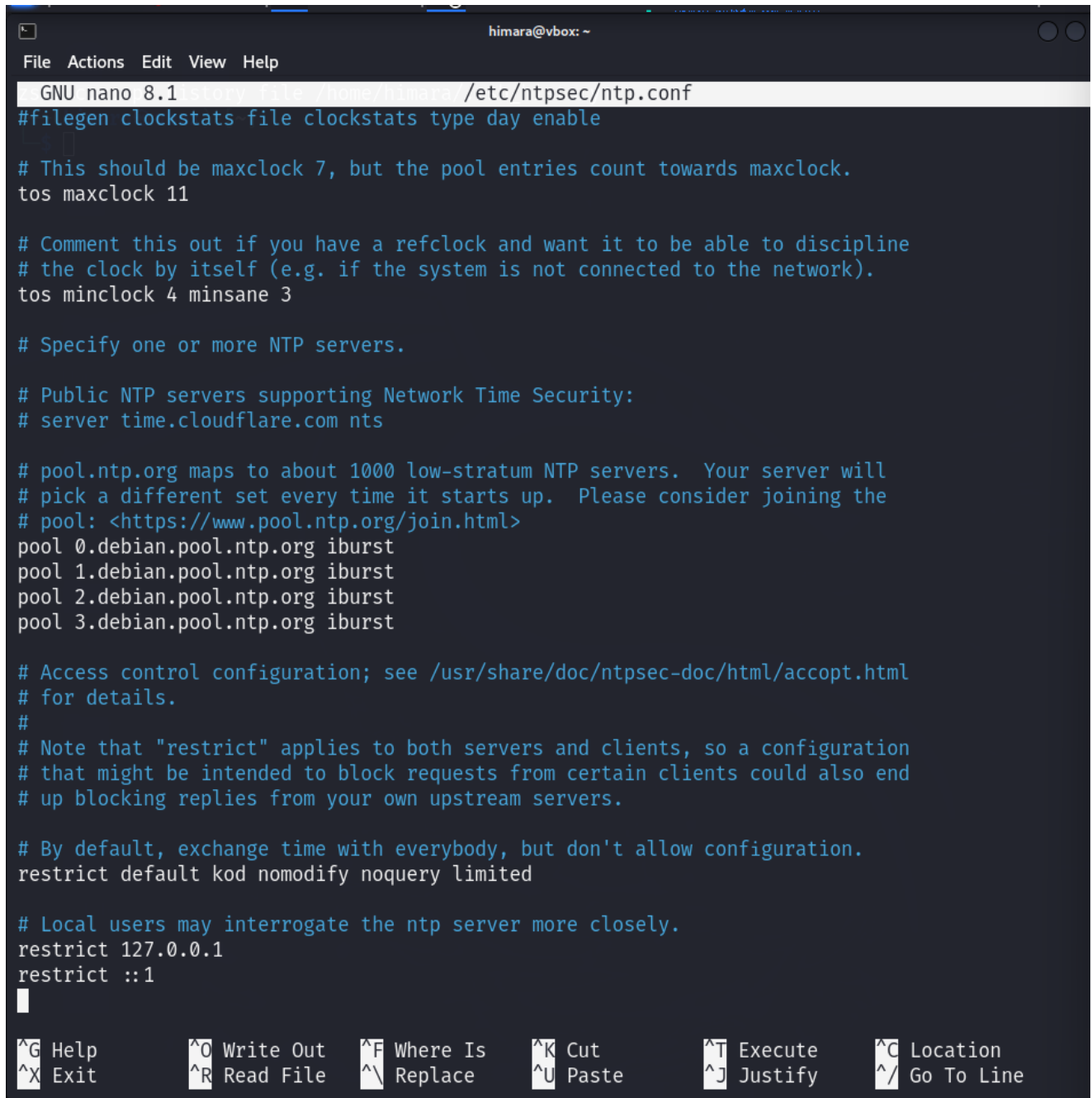
Suggested packages:
certbot ntpsec-doc ntpsec-ntpviz

Summary:
Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 2121
Download size: 459 kB
Space needed: 1364 kB / 1162 MB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 python3-ntp amd64 1.2.3+dfsg1-3 [90.3 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 ntpsec amd64 1.2.3+dfsg1-3 [345 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 ntp all 1:4.2.8p15+dfsg-2~1.2.3+dfsg1-3 [23.4 kB]
Fetched 459 kB in 1s (312 kB/s)
Selecting previously unselected package python3-ntp.
(Reading database ... 396626 files and directories currently installed.)
Preparing to unpack .../python3-ntp_1.2.3+dfsg1-3_amd64.deb ...
Unpacking python3-ntp (1.2.3+dfsg1-3) ...
Selecting previously unselected package ntpsec.
Preparing to unpack .../ntpsec_1.2.3+dfsg1-3_amd64.deb ...
Unpacking ntpsec (1.2.3+dfsg1-3) ...
Selecting previously unselected package ntp.
Preparing to unpack .../ntp_1%3a4.2.8p15+dfsg-2~1.2.3+dfsg1-3_all.deb ...
Unpacking ntp (1:4.2.8p15+dfsg-2~1.2.3+dfsg1-3) ...
Setting up python3-ntp (1.2.3+dfsg1-3) ...
Setting up ntpsec (1.2.3+dfsg1-3) ...
update-rc.d: We have no instructions for the ntpsec init script.
update-rc.d: It looks like a network service, we disable it.
Created symlink '/etc/systemd/system/timers.target.wants/ntpsec-rotate-stats.timer' -> '/usr/lib/systemd/system/ntpsec-rotate-stats.timer'.
Created symlink '/etc/systemd/system/network-pre.target.wants/ntpsec-systemd-netif.path' -> '/usr/lib/systemd/system/ntpsec-systemd-netif.path'.
ntpsec.service is a disabled or a static unit, not starting it.
```

3. Configure NTP server

Use thi command “**sudo nano /etc/ntpsec/ntp.conf** ” you can get a /etc/ntpsec/ntp.conf file and you can edit the main configuration file for NTP. After open the file we can replace or modify the lines with the NTP server you want to use.



```
himara@vbox: ~
File Actions Edit View Help
GNU nano 8.1 history file /home/himara /etc/ntpsec/ntp.conf
#filegen clockstats file clockstats type day enable

# This should be maxclock 7, but the pool entries count towards maxclock.
tos maxclock 11

# Comment this out if you have a refclock and want it to be able to discipline
# the clock by itself (e.g. if the system is not connected to the network).
tos minclock 4 minsane 3

# Specify one or more NTP servers.

# Public NTP servers supporting Network Time Security:
# server time.cloudflare.com nts

# pool.ntp.org maps to about 1000 low-stratum NTP servers. Your server will
# pick a different set every time it starts up. Please consider joining the
# pool: <https://www.pool.ntp.org/join.html>
pool 0.debian.pool.ntp.org iburst
pool 1.debian.pool.ntp.org iburst
pool 2.debian.pool.ntp.org iburst
pool 3.debian.pool.ntp.org iburst

# Access control configuration; see /usr/share/doc/ntpsec-doc/html/accopt.html
# for details.
#
# Note that "restrict" applies to both servers and clients, so a configuration
# that might be intended to block requests from certain clients could also end
# up blocking replies from your own upstream servers.

# By default, exchange time with everybody, but don't allow configuration.
restrict default kod nomodify noquery limited

# Local users may interrogate the ntp server more closely.
restrict 127.0.0.1
restrict ::1

```

^G Help **^O Write Out** **^F Where Is** **^K Cut** **^T Execute** **^C Location**
^X Exit **^R Read File** **^_ Replace** **^U Paste** **^J Justify** **^_/ Go To Line**

4. Star and enable NTP

“sudo systemctl start ntpsec” Using this command now you can start and enable the NTP.

```
(himara@vbox)-[~]
$ sudo systemctl start ntpsec

(himara@vbox)-[~]
$ sudo systemctl enable ntpsec
Synchronizing state of ntpsec.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ntpsec
Created symlink '/etc/systemd/system/ntp.service' → '/usr/lib/systemd/system/ntpsec.service'.
Created symlink '/etc/systemd/system/ntpd.service' → '/usr/lib/systemd/system/ntpsec.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ntpsec.service' → '/usr/lib/systemd/system/ntpsec.service'.

(himara@vbox)-[~]
$
```

5. View the status of the NTP

```
(himara@vbox)-[~]
$ sudo systemctl status ntpsec
● ntpsec.service - Network Time Service
   Loaded: loaded (/usr/lib/systemd/system/ntpsec.service; enabled; preset: disabled)
   Active: active (running) since Thu 2025-05-01 16:10:49 EDT; 1min 16s ago
     Invocation: 4592f6bf6d3747e3ac1175ee97326928
       Docs: man:ntpd(8)
    Main PID: 7643 (ntpd)
       Tasks: 1 (limit: 4548)
      Memory: 13M (peak: 13.5M)
         CPU: 97ms
    CGroup: /system.slice/ntpsec.service
            └─7643 /usr/sbin/ntpd -p /run/ntpd.pid -c /etc/ntpsec/ntp.conf -g -N -u ntpsec:ntpsec

May 01 16:10:52 vbox ntpd[7643]: DNS: Pool taking: 2606:4700:f1::1
May 01 16:10:52 vbox ntpd[7643]: DNS: dns_take_status: 2.debian.pool.ntp.org⇒good, 8
May 01 16:10:53 vbox ntpd[7643]: DNS: dns_probe: 3.debian.pool.ntp.org, cast_flags:8, flags:101
May 01 16:10:53 vbox ntpd[7643]: DNS: dns_check: processing 3.debian.pool.ntp.org, 8, 101
May 01 16:10:53 vbox ntpd[7643]: DNS: Pool skipping: 222.165.180.134
May 01 16:10:53 vbox ntpd[7643]: DNS: Pool skipping: 162.159.200.123
May 01 16:10:53 vbox ntpd[7643]: DNS: Pool skipping: 162.159.200.1
May 01 16:10:53 vbox ntpd[7643]: DNS: dns_take_status: 3.debian.pool.ntp.org⇒good, 8
May 01 16:10:58 vbox ntpd[7643]: CLOCK: time stepped by 1.534429
May 01 16:10:58 vbox ntpd[7643]: INIT: MRU 10922 entries, 13 hash bits, 65536 bytes

(himara@vbox)-[~]
$
```


6. Verify Time Synchronization

In Linux, the `ntpq -p` command is used to display a list of NTP peers that the local machine is synced with and to query Network Time Protocol (NTP) servers for their current status.

```
(himara@vbox)-[~]
$ ntpq -p
remote                                refid      st t when poll reach  delay  offset  jitter
-----
0.debian.pool.ntp.org                .POOL.     16 p   - 256    0   0.0000   0.0000   0.0000
1.debian.pool.ntp.org                .POOL.     16 p   - 256    0   0.0000   0.0000   0.0000
2.debian.pool.ntp.org                .POOL.     16 p   - 256    0   0.0000   0.0000   0.0000
3.debian.pool.ntp.org                .POOL.     16 p   - 256    0   0.0000   0.0000   0.0000
+time.cloudflare.com                10.154.8.7  3 u    5  64    7  67.1583   3.9268   8.6179
*ntp.sltidc.lk                      216.239.35.4 2 u    5  64    7  21.4303  -3.1807  99.4783
+time.cloudflare.com                10.229.8.116 3 u    5  64    7  52.2257  -3.5106   5.8641
+time.cloudflare.com                10.228.8.4   3 u    -  64   17  66.9259   4.8640 108.0435
+time.cloudflare.com                10.156.9.195 3 u   63  64    7  66.2598   4.6179   6.5976

(himara@vbox)-[~]
$
```

The output of `timedatectl status` shows that the system's local time is Eastern Daylight Time (EDT) and that the system is synchronized with NTP. The system is using UTC as the hardware clock (RTC), and the time zone is correctly set to America/New_York.

```
(himara@vbox)-[~]
$ timedatectl status
          Local time: Thu 2025-05-01 16:16:45 EDT
          Universal time: Thu 2025-05-01 20:16:45 UTC
             RTC time: Thu 2025-05-01 20:16:43
          Time zone: America/New_York (EDT, -0400)
System clock synchronized: yes
           NTP service: active
          RTC in local TZ: no

(himara@vbox)-[~]
$
```

IV. Need and use of the network services in administration

a) DHCP

- **Need:** DHCP is needed in order to have devices automatically configured in a network. Without DHCP, each device (computers, phones, printers) would have to be statically configured with an IP. Not only is this a laborious process, but it is also prone to errors. DHCP allows it to be simplified to such a degree so that it reduces administrative work and avoids conflicts when different devices would have the same IP.
- **Use:** They make devices part of a network and are configured with an IP address, subnet mask, etc., by automatically assigning them. DHCP ensures in a large network that each device will have a unique IP address without manual configuration by an administrator.

b) DNS

- **Need:** DNS is the internet "phonebook." It maps human-readable addresses (i.e., `www.example.com`) to computer-readable IP addresses so other devices can locate each other on a network. Without DNS, people would have to remember and type numeric IP addresses to access a website, making it impossible.
- **Use:** DNS facilitates internet browsing for users. DNS servers convert an entered website address in a browser into a relevant IP server address on which the site is hosted. DNS is involved in services for an internal network as well as browsing on the external internet.

c) NTP

- **Need:** Accurate timekeeping is crucial to a great deal of network functionality, including such things as event logging, data integrity and synchronizing equipment. Without synchronizing equipment to a universal time, events like automated backups, data transfer or security logs are undependable and in disarray.
- **Use:** NTP synchronizes all devices in a network simultaneously without errors and discrepancies. NTP synchronizes computer clocks to an extremely precise time source such as an atomic clock or GPS so that devices will be accurate.

3. Security and other servers

1. Shell Scripting

Shell Scripting Basics:

- If-else statement



```
himara@vbox: ~/SNP_Lab
File Actions Edit View Help
zsh: corrupt history file /home/himara/.zsh_history
(himara@vbox)-[~/SNP_Lab]
$ nano script.sh

(himara@vbox)-[~/SNP_Lab]
$ ./script.sh
Please enter your name:
Himara
Welcome back Himara

(himara@vbox)-[~/SNP_Lab]
$ ./script.sh
Please enter your name:
Vimansa
Invalid Username

(himara@vbox)-[~/SNP_Lab]
$
```

```
File Actions Edit View Help
GNU nano 8.1 history file /home/himara/.zsh_history script.sh
#!/bin/bash
echo "Please enter your name:"
read NAME

if [ "$NAME" = "Himara" ];
then
    echo "Welcome back Himara"

else
    echo "Invalid Username"

fi
```

- For loop

```
himara@vbox: ~/SNP_Lab
File Actions Edit View Help
zsh: corrupt_history file /home/himara/.zsh_history
(himara@vbox)-[~/SNP_Lab]
$ nano loop_example.sh

(himara@vbox)-[~/SNP_Lab]
$ chmod +x loop_example.sh

(himara@vbox)-[~/SNP_Lab]
$ ./loop_example.sh
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5

(himara@vbox)-[~/SNP_Lab]
$
```

```
himara@vbox: ~/SNP_Lab
File Actions Edit View Help
GNU nano 8.1 history file /home/himara/function_example.sh *
#!/bin/bash
#
print_numbers() {
    for i in {1..5}
    do
        echo "Number: $i"
    done
}

print_numbers
```

- While loop

```
(himara@vbox)-[~/SNP_Lab]
$ nano while_example.sh

(himara@vbox)-[~/SNP_Lab]
$ chmod +x while_example.sh

(himara@vbox)-[~/SNP_Lab]
$ ./while_example.sh
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5

(himara@vbox)-[~/SNP_Lab]
$
```

```
himara@vbox: ~/SNP_Lab
File Actions Edit View Help
GNU nano 8.1 history file /home/himara/ while_example.sh *
#!/bin/bash
i=1
while [ $i -le 5 ]
do
    echo "Number: $i"
    ((i++))
done
```

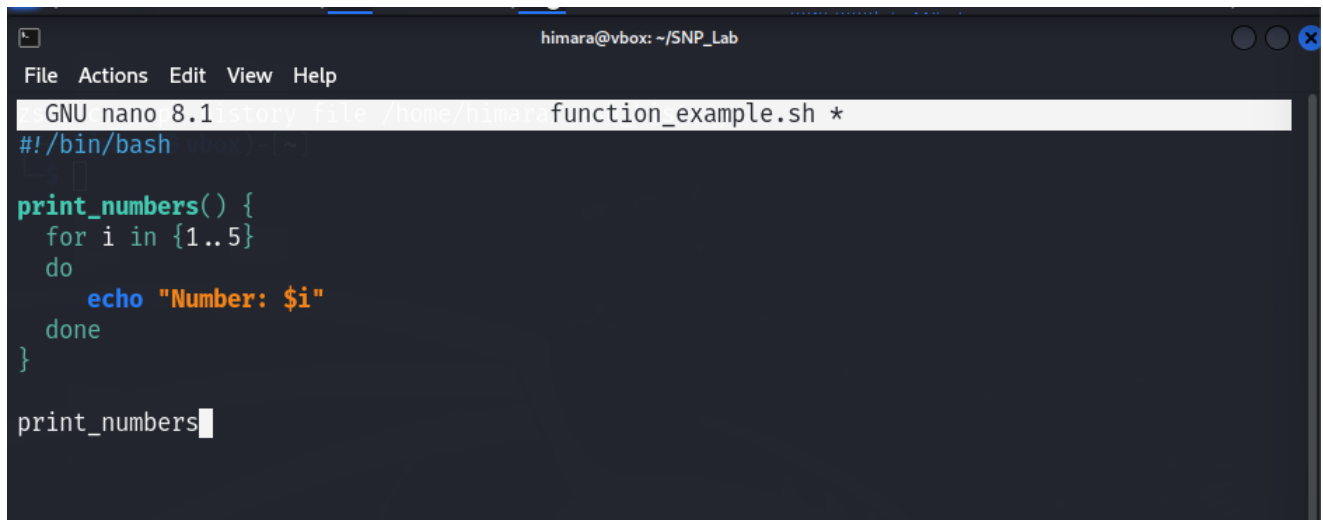
- Function

```
himara@vbox: ~/SNP_Lab
File Actions Edit View Help
zsh: corrupt history file /home/himara/.zsh_history
(himara@vbox)-[~/SNP_Lab]
$ nano function_example.sh

(himara@vbox)-[~/SNP_Lab]
$ chmod +x function_example.sh

(himara@vbox)-[~/SNP_Lab]
$ ./function_example.sh
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5

(himara@vbox)-[~/SNP_Lab]
$
```

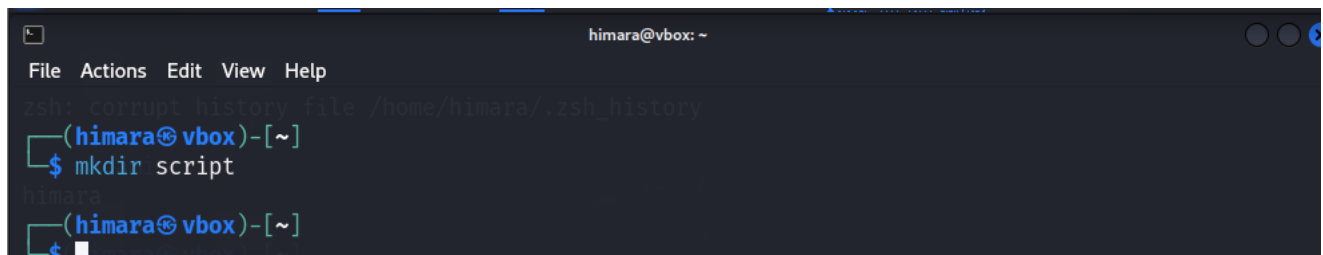


```
himara@vbox: ~/SNP_Lab
File Actions Edit View Help
GNU nano 8.1 history file /home/himara/function_example.sh *
#!/bin/bash
#
#
print_numbers() {
  for i in {1..5}
  do
    echo "Number: $i"
  done
}

print_numbers
```

Task 01

1. Create a new directory in /home/himara with the name **script**.



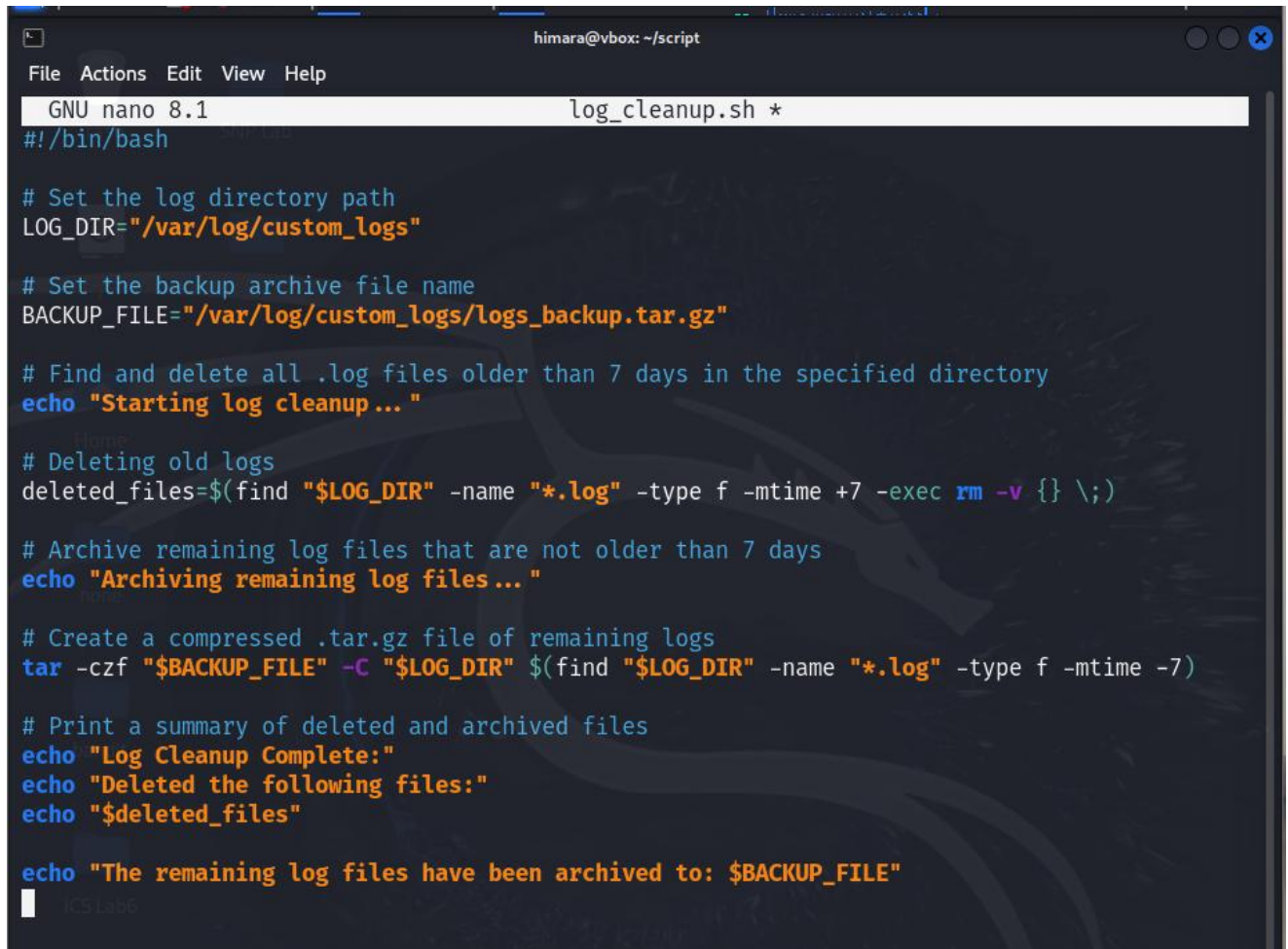
```
himara@vbox: ~
File Actions Edit View Help
zsh: corrupt history file /home/himara/.zsh_history
(himara@vbox)-[~]
$ mkdir script
himara
(himara@vbox)-[~]
$
```

2. You can use a text editor **nano** to create the script. You can create a nano log_cleanup.sh text file.



```
(himara@vbox)-[~/script]
$ nano log_cleanup.sh
```

3. Write the script

A screenshot of a terminal window titled 'himara@vbox: ~/script'. The window shows the GNU nano 8.1 text editor editing a file named 'log_cleanup.sh'. The script content is as follows:

```
#!/bin/bash

# Set the log directory path
LOG_DIR="/var/log/custom_logs"

# Set the backup archive file name
BACKUP_FILE="/var/log/custom_logs/logs_backup.tar.gz"

# Find and delete all .log files older than 7 days in the specified directory
echo "Starting log cleanup..."

# Deleting old logs
deleted_files=$(find "$LOG_DIR" -name "*.log" -type f -mtime +7 -exec rm -v {} \;)

# Archive remaining log files that are not older than 7 days
echo "Archiving remaining log files..."

# Create a compressed .tar.gz file of remaining logs
tar -czf "$BACKUP_FILE" -C "$LOG_DIR" $(find "$LOG_DIR" -name "*.log" -type f -mtime -7)

# Print a summary of deleted and archived files
echo "Log Cleanup Complete:"
echo "Deleted the following files:"
echo "$deleted_files"

echo "The remaining log files have been archived to: $BACKUP_FILE"
```

- **LOG_DIR="/var/log/custom_logs"** = Set the log directory path.
- **BACKUP_FILE="/var/log/custom_logs/logs_backup.tar.gz"** = Set the backup archive file name.
- **echo "Starting log cleanup..."** = Find and delete all .log files older than 7 days in the specified directory.
- **deleted_files=\$(find "\$LOG_DIR" -name "*.log" -type f -mtime +7 -exec rm -v {} \;)** = Deleting old logs.
- **echo "Archiving remaining log files..."** = Archive remaining log files that are not older than 7 days.
- **tar -czf "\$BACKUP_FILE" -C "\$LOG_DIR" \$(find "\$LOG_DIR" -name "*.log" -type f -mtime -7)** = Create a compressed .tar.gz file of remaining logs.

- **echo "Log Cleanup Complete:"**
- **echo "Deleted the following files:"**
- **echo "\$deleted_files"**
- **echo "The remaining log files have been archived to: \$BACKUP_FILE"**

print a summary of deleted and archived files.

4. Make the script executable

To execute the script, you should provide it with executable permissions. Run the following command in the terminal. This will render the script executable.

```
(himara@vbox)-[~/script]  
$ chmod +x log_cleanup.sh
```

5. Test the Script

```
(himara@vbox)-[~/script]  
$ ./log_cleanup.sh  
  
Starting log cleanup...  
Archiving remaining log files...  
tar: Cowardly refusing to create an empty archive  
Try 'tar --help' or 'tar --usage' for more information.  
Log Cleanup Complete:  
Deleted the following files:  
  
The remaining log files have been archived to: /var/log/custom_logs/logs_backup.tar.gz
```

6. Set Up a Cron Job

You must add this script to your cron tasks in order to have it run automatically every Sunday at midnight. Open the crontab file by “**crontab -e**” running thi code and add the following line to run the script every Sunday at 12:00 AM (midnight).

```
(himara@vbox)-[~/script]
$ crontab -e
```

```
no crontab for himara - using an empty one
crontab: installing new crontab
```

```
himara@vbox: ~/script
File Actions Edit View Help
GNU nano 8.1 story file /home/himara/tmp/crontab.8mNnwU/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 0 * * SUN /home/himara/log_cleanup.sh >> /home/himara/log_cleanup.log 2>&1
```

Task 02

SSH (Secure Shell):

1. Update and Install the SSH server package

```
(himara@vbox)-[~]  
$ sudo apt update  
  
[sudo] password for himara:  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
2135 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

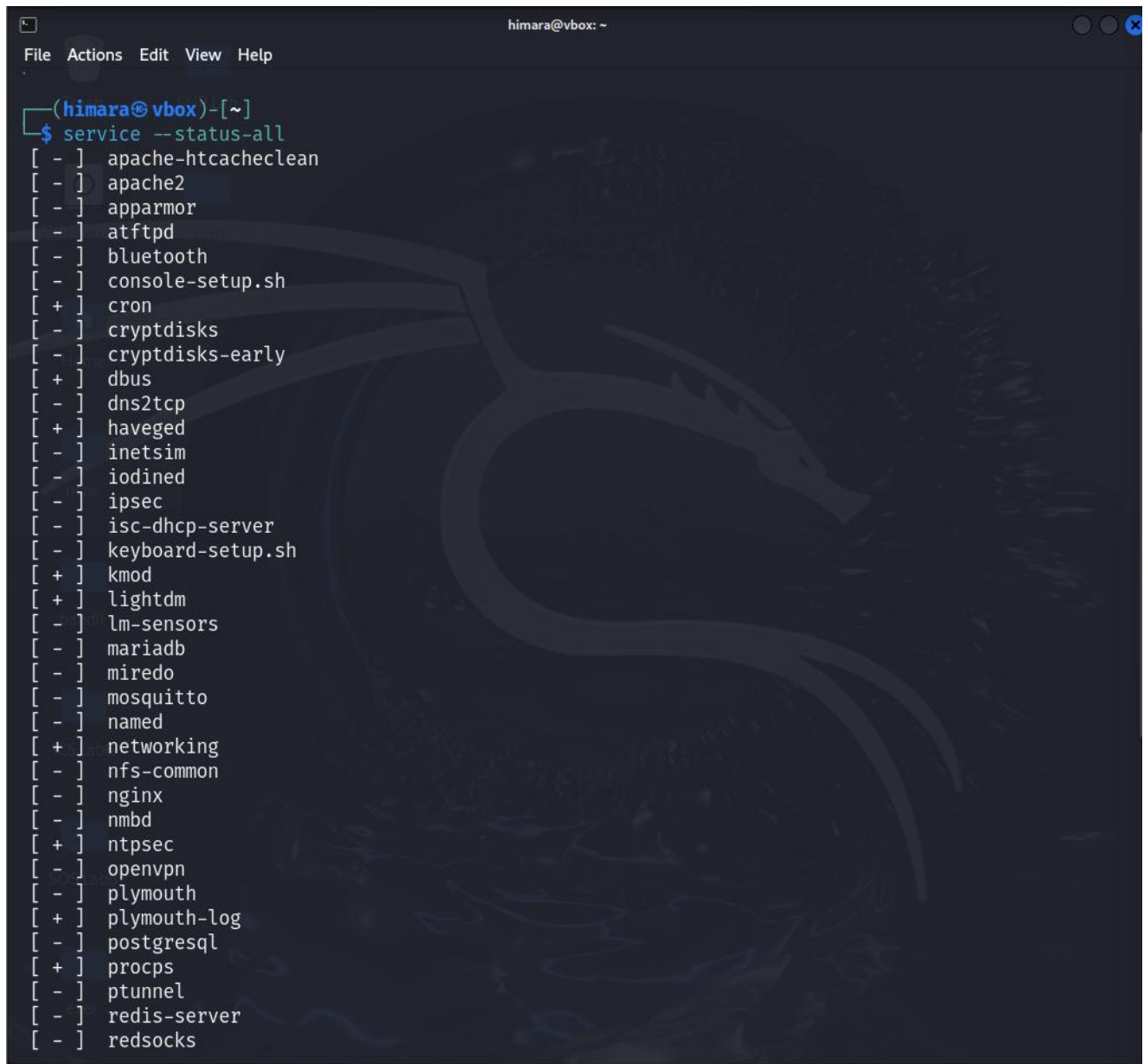
```
(himara@vbox)-[~]  
$ sudo apt install openssh-server  
  
openssh-server is already the newest version (1:9.9p2-2).  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2135
```

2. Verify the SSH service is running

Check whether if the SSH service is running properly.

```
(himara@vbox)-[~]  
$ sudo systemctl status ssh  
  
● ssh.service - OpenBSD Secure Shell server  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)  
   Active: active (running) since Sun 2025-05-04 09:51:18 EDT; 3min 52s ago  
 Invocation: ed1b46a8df2e4bc0923f7ffa28cb597e  
    Docs: man:sshd(8)  
          man:sshd_config(5)  
  Process: 838 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)  
 Main PID: 857 (sshd)  
    Tasks: 1 (limit: 4548)  
  Memory: 2.1M (peak: 2.3M)  
     CPU: 92ms  
   CGroup: /system.slice/ssh.service  
           └─857 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
May 04 09:51:18 vbox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...  
May 04 09:51:18 vbox sshd[857]: Server listening on 0.0.0.0 port 22.  
May 04 09:51:18 vbox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.  
May 04 09:51:18 vbox sshd[857]: Server listening on :: port 22.
```

- After installing and enabling SSH, now we check the status of all available services on your system.
- It provides a quick overview of whether services are running, stopped, or inactive.
- According to the below image the SSH service now active, which is shown by ‘+’ mark.

A terminal window titled 'himara@vbox: ~' with a menu bar (File, Actions, Edit, View, Help). The prompt is '(himara@vbox)-[~]'. The command '\$ service --status-all' has been executed, resulting in a list of services with status indicators in brackets. The 'cron' service is marked with a '+', indicating it is active. A large, faint dragon watermark is visible in the background of the terminal.

```
(himara@vbox)-[~]
$ service --status-all
[ - ] apache-htcacheclean
[ - ] apache2
[ - ] apparmor
[ - ] atftpd
[ - ] bluetooth
[ - ] console-setup.sh
[ + ] cron
[ - ] cryptdisks
[ - ] cryptdisks-early
[ + ] dbus
[ - ] dns2tcp
[ + ] haveged
[ - ] inetsim
[ - ] iodined
[ - ] ipsec
[ - ] isc-dhcp-server
[ - ] keyboard-setup.sh
[ + ] kmod
[ + ] lightdm
[ - ] lm-sensors
[ - ] mariadb
[ - ] miredo
[ - ] mosquitto
[ - ] named
[ + ] networking
[ - ] nfs-common
[ - ] nginx
[ - ] nmbd
[ + ] ntpsec
[ - ] openvpn
[ - ] plymouth
[ + ] plymouth-log
[ - ] postgresql
[ + ] procps
[ - ] ptunnel
[ - ] redis-server
[ - ] redsocks
```

```
[ - ] rpcbind
[ - ] rsync
[ - ] rwhod
[ - ] saned
[ - ] screen-cleanup
[ - ] selinux-autorelabel
[ - ] smartmontools
[ - ] smbd
[ - ] snmpd
[ - ] speech-dispatcher
[ - ] ssh
[ - ] sslh
[ - ] stunnel4
[ - ] sudo
[ - ] sysstat
[ + ] ufw
[ + ] virtualbox-guest-utils
[ - ] x11-common
```

```
(himara@vbox)-[~]
$
```

3. Enable SSH service to start on boot

```
(himara@vbox)-[~]
$ sudo systemctl enable ssh
```

```
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sy
sv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
```

4. Edit the SSH configuration file

- The SSH configuration file is located at `/etc/ssh/sshd_config`.

```
(himara@vbox)-[~]
$ sudo nano /etc/ssh/sshd_config
```

```
(himara@vbox)-[~]
$
```

After accessing the this file you can do the following modifications:

- Port = 2255
- PermitRootLogin = no
- PasswordAuthentication = Set this to 'no' if you want to enforce key based authentication.
- AllowUsers: Specify which users are allowed to connect via SSH.

```
himara@vbox: ~
File Actions Edit View Help
GNU nano 8.1 /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 2255
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

[ Read 122 lines ]
^G Help      ^O Write Out  ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

5. Restart the service

After making this changes restart the SSH service.

```
(himara@vbox)-[~]
$ sudo systemctl restart ssh

(himara@vbox)-[~]
$
```

6. Allow SSH through the firewall

```
(himara@vbox)-[~]
$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)

(himara@vbox)-[~]
$ sudo ufw allow 2255/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)

(himara@vbox)-[~]
$ sudo ufw enable
Firewall is active and enabled on system startup

(himara@vbox)-[~]
$ sudo ufw status
Status: active
```

To	Action	From
--	---	---
22/tcp	ALLOW	Anywhere
2255/tcp	ALLOW	Anywhere
67/udp	ALLOW	Anywhere
53/udp	ALLOW	Anywhere
53/tcp	ALLOW	Anywhere
22/tcp (v6)	ALLOW	Anywhere (v6)
2255/tcp (v6)	ALLOW	Anywhere (v6)
67/udp (v6)	ALLOW	Anywhere (v6)
53/udp (v6)	ALLOW	Anywhere (v6)
53/tcp (v6)	ALLOW	Anywhere (v6)

7. Locating the IP address of the server computer

```
(himara@vbox)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fe0b:111a prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:0b:11:1a txqueuelen 1000 (Ethernet)
    RX packets 8 bytes 2634 (2.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 20 bytes 3520 (3.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 14 bytes 780 (780.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14 bytes 780 (780.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```


8. Connect to the Linux Machine from Another Computer Using SSH

Open a terminal on the remote machine and run the following command:

```
(himara@vbox)-[~]  
$ ssh himara@192.168.56.103 -p 2255  
Linux vbox 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64  
  
The programs included with the Kali GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Sat May 3 16:06:56 2025 from 192.168.56.103  
zsh: corrupt history file /home/himara/.zsh_history
```

Task 03

iptables and ACLs

1. Find the IP ranges for Social Media Websites
 - Facebook: 157.240.0.0/16 (primary IP range for Facebook)
 - Instagram: 31.13.64.0/18 (Instagram is owned by Facebook, so the IP ranges overlap)
 - Twitter: 199.59.148.0/22 (Twitter's IP range)
2. Add iptables Rules to Block Social Media Websites

Open your VM terminal and run the following iptables command to block the social media IP.

```
File Actions Edit View Help  
zsh: corrupt history file /home/himara/.zsh_history  
(himara@vbox)-[~]  
$ # Block Facebook IP range  
sudo iptables -A OUTPUT -d 157.240.0.0/16 -j REJECT # Facebook  
  
# Block Instagram IP range (same as Facebook's IP ranges)  
sudo iptables -A OUTPUT -d 31.13.64.0/18 -j REJECT # Instagram  
  
# Block Twitter IP range  
sudo iptables -A OUTPUT -d 199.59.148.0/22 -j REJECT # Twitter  
  
[sudo] password for himara:  
(himara@vbox)-[~]  
$
```


3. Save the iptables Rules

```
(himara@vbox)-[~]  
$ sudo iptables-save > /etc/iptables/rules.v4
```

4. Allow Only Secure Web Browsing (HTTPS)

Block HTTP (Port 80) Traffic

```
(himara@vbox)-[~]  
$ sudo iptables -A OUTPUT -p tcp --dport 80 -j REJECT # Block HTTP (port 80)
```

Allow HTTPS (Port 443) Traffic

```
(himara@vbox)-[~]  
$ sudo iptables -A OUTPUT -p tcp --dport 443 -j ACCEPT # Allow HTTPS (port 443)
```

5. Install of the iptables persistent

```
(himara@vbox)-[~]  
$ sudo apt install iptables-persistent  
  
iptables-persistent is already the newest version (1.0.23).  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2123
```

6. Save the iptables Rules

```
(himara@vbox)-[~]  
$ sudo iptables-save > /etc/iptables/rules.v4
```

7. Verify the Configuration

```
(himara@vbox)-[~]
$ sudo iptables -L

Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
REJECT     all  --  anywhere             157.240.0.0/16      reject-with icmp-port-unreachable
REJECT     all  --  anywhere             31.13.64.0/18       reject-with icmp-port-unreachable
REJECT     all  --  anywhere             199.59.148.0/22     reject-with icmp-port-unreachable
REJECT     tcp  --  anywhere             anywhere            tcp dpt:http reject-with icmp-port-unreachable
ACCEPT     tcp  --  anywhere             anywhere            tcp dpt:https
```

Task 04

Web Server (Apache/Nginx)

1. Install Apache2

```
(himara@vbox)-[~]
$ sudo apt-get install apache2
[sudo] password for himara:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.63-1).
apache2 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 2127 not upgraded.
```

2. Start the apache2

```
(himara@vbox)-[~]
$ sudo systemctl start apache2
sudo systemctl enable apache2

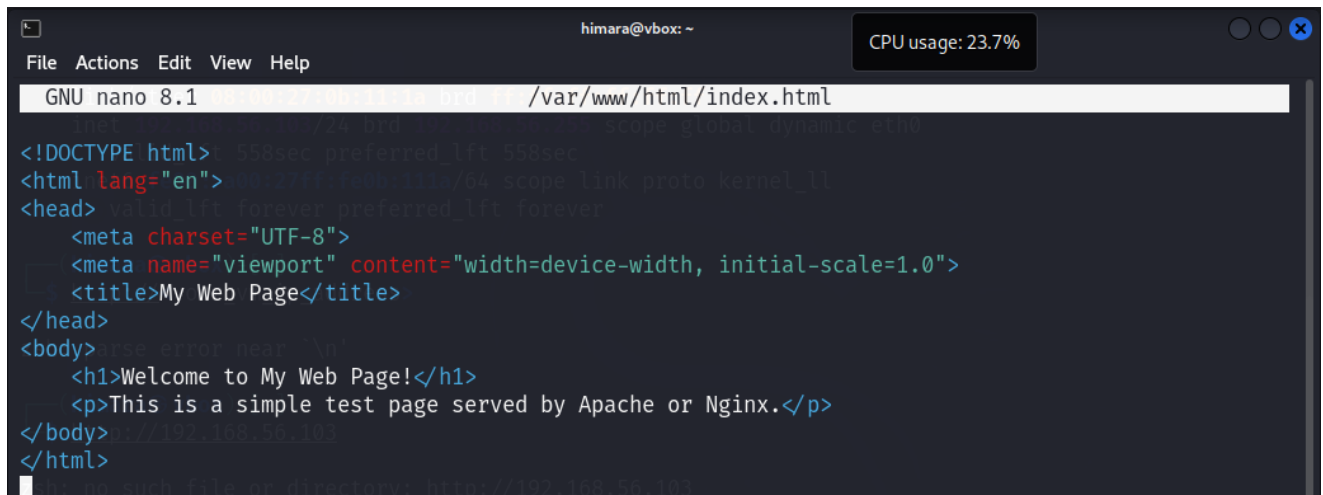
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

3. Check if Apache is running

```
(himara@vbox)-[~]
$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Tue 2025-05-06 12:58:17 EDT; 1h 4min ago
     Invocation: 4485e797a54945ea861bd5018cc4fb3c
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 857 (apache2)
      Tasks: 6 (limit: 4548)
     Memory: 20.6M (peak: 21.1M)
        CPU: 809ms
    CGroup: /system.slice/apache2.service
            └─857 /usr/sbin/apache2 -k start
              └─861 /usr/sbin/apache2 -k start
                └─862 /usr/sbin/apache2 -k start
                  └─863 /usr/sbin/apache2 -k start
                    └─864 /usr/sbin/apache2 -k start
                      └─865 /usr/sbin/apache2 -k start

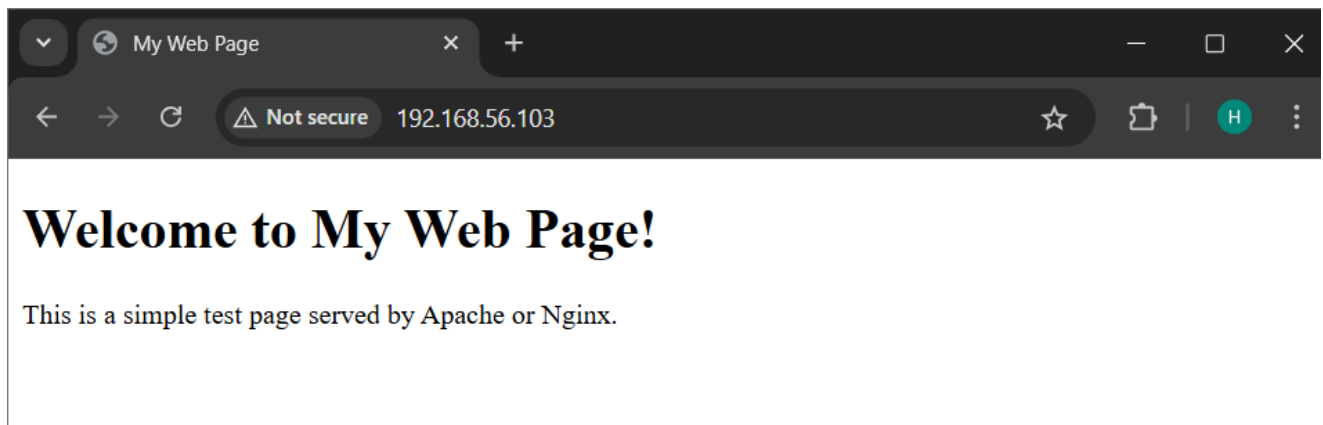
May 06 12:58:17 vbox systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 06 12:58:17 vbox systemd[1]: Started apache2.service - The Apache HTTP Server.
```

4. Create an **index.html** file in the web server's root directory



```
himara@vbox: ~
File Actions Edit View Help
GNU nano 8.1 /var/www/html/index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Web Page</title>
</head>
<body>
  <h1>Welcome to My Web Page!</h1>
  <p>This is a simple test page served by Apache or Nginx.</p>
</body>
</html>
```

5. Access the page from another machine



Task 05

Email Server (postfix/Exim)

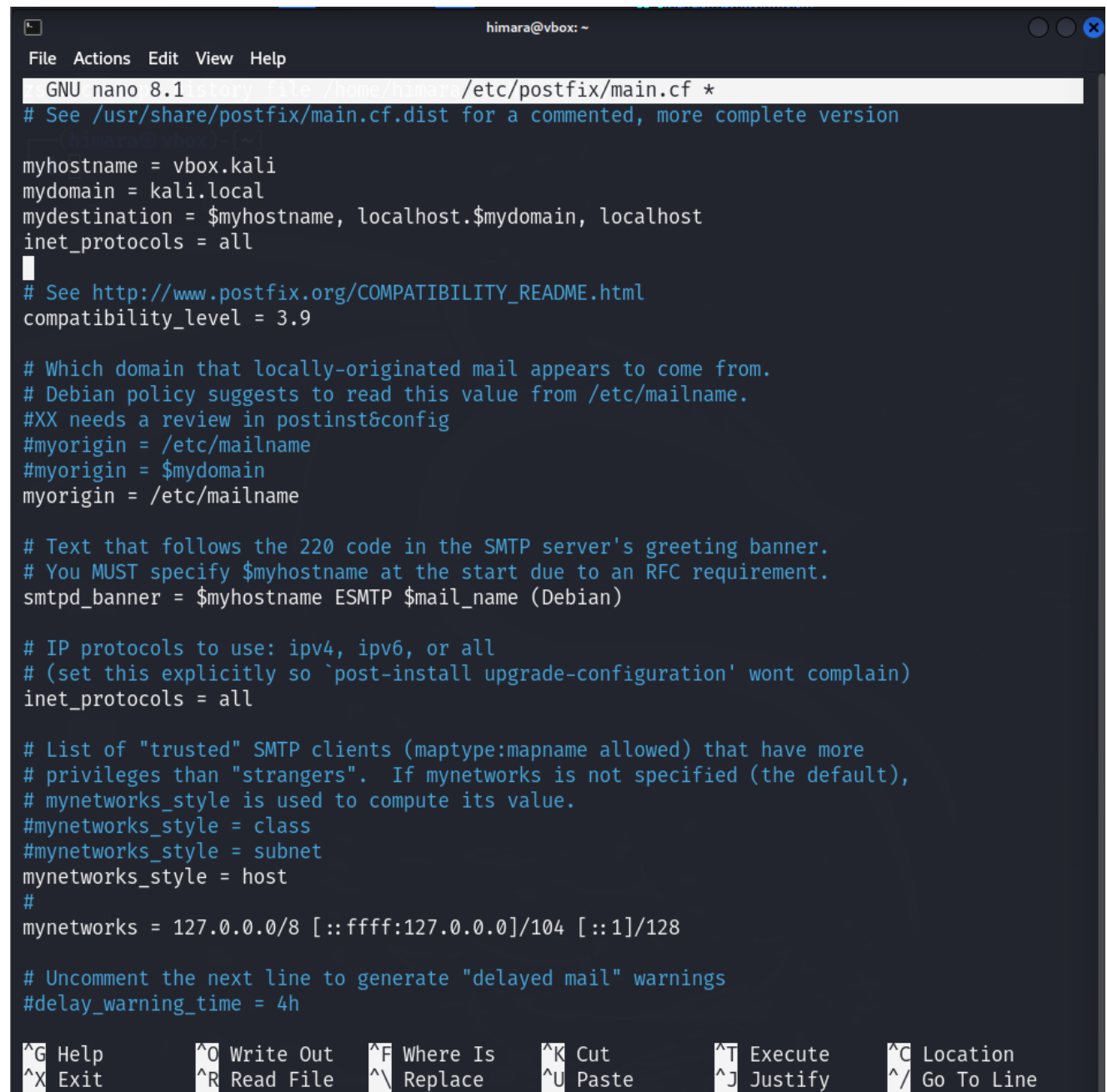
1. Install Postfix

```
himara@vbox: ~  
File Actions Edit View Help  
(himara@vbox)-[~] file=/home/himara/.zsh_history  
$ sudo apt install postfix  
  
Upgrading:  
  icu-devtools  libicu-dev  
  
Installing:  
  postfix  
  
Installing dependencies:  
  libicu76  libtlsrpt0  
  
Suggested packages:  
  postfix-cdb      postfix-lmdb      postfix-mysql    postfix-sqlite    sasl2-bin  
  postfix-doc      postfix-mta-sts-resolver  postfix-pcre     procmail          | dovecot-common  
  postfix-ldap     postfix-mongodb   postfix-pgsql    resolvconf        ufw  
  
REMOVING:  
  exim4-base  exim4-config  exim4-daemon-light  
  
Summary:  
  Upgrading: 2, Installing: 3, Removing: 3, Not Upgrading: 2124  
  Download size: 1603 kB / 22.3 MB  
  Space needed: 40.3 MB / 822 MB available  
  
Continue? [Y/n] y  
Get:1 http://kali.download/kali kali-rolling/main amd64 postfix amd64 3.10.2-1 [1593 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 libtlsrpt0 amd64 0.5.0rc1-2 [9588 B]  
Fetched 1603 kB in 6s (288 kB/s)  
Preconfiguring packages ...  
dpkg: exim4-daemon-light: dependency problems, but removing anyway as you requested:  
  bsd-mailx depends on default-mta | mail-transport-agent; however:  
    Package default-mta is not installed.  
    Package exim4-daemon-light which provides default-mta is to be removed.  
    Package mail-transport-agent is not installed.  
    Package exim4-daemon-light which provides mail-transport-agent is to be removed.  
  bsd-mailx depends on default-mta | mail-transport-agent; however:  
    Package default-mta is not installed.  
    Package exim4-daemon-light which provides default-mta is to be removed.  
    Package mail-transport-agent is not installed.
```

Configure Postfix:

2. Edit the Postfix configuration file

```
(himara@vbox)-[~]  
$ sudo nano /etc/postfix/main.cf
```



```
GNU nano 8.1 /etc/postfix/main.cf *  
# See /usr/share/postfix/main.cf.dist for a commented, more complete version  
  
myhostname = vbox.kali  
mydomain = kali.local  
mydestination = $myhostname, localhost.$mydomain, localhost  
inet_protocols = all  
# See http://www.postfix.org/COMPATIBILITY_README.html  
compatibility_level = 3.9  
  
# Which domain that locally-originated mail appears to come from.  
# Debian policy suggests to read this value from /etc/mailname.  
#XX needs a review in postinst&config  
#myorigin = /etc/mailname  
#myorigin = $mydomain  
myorigin = /etc/mailname  
  
# Text that follows the 220 code in the SMTP server's greeting banner.  
# You MUST specify $myhostname at the start due to an RFC requirement.  
smtpd_banner = $myhostname ESMTP $mail_name (Debian)  
  
# IP protocols to use: ipv4, ipv6, or all  
# (set this explicitly so `post-install upgrade-configuration' wont complain)  
inet_protocols = all  
  
# List of "trusted" SMTP clients (maptype:mapname allowed) that have more  
# privileges than "strangers". If mynetworks is not specified (the default),  
# mynetworks_style is used to compute its value.  
#mynetworks_style = class  
#mynetworks_style = subnet  
mynetworks_style = host  
#  
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128  
  
# Uncomment the next line to generate "delayed mail" warnings  
#delay_warning_time = 4h  
  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

3. Restart Postfix

```
(himara@vbox)-[~]  
$ sudo systemctl restart postfix
```

4. Test Postfix Installation

Install the mail command

```
himara@vbox: ~  
File Actions Edit View Help  
zsh: corrupt history file /home/himara/.zsh_history  
(himara@vbox)-[~]  
$ sudo apt install mailutils  
  
Upgrading:  
  libltdl7  
  
Installing:  
  mailutils  
  
Installing dependencies:  
  gsasl-common  libgsasl18  libmailutils9t64  mailutils-common  
  guile-3.0-libs  libgssglue1  libntlm0  
  
Suggested packages:  
  mailutils-mh  mailutils-doc  
  
Summary:  
  Upgrading: 1, Installing: 8, Removing: 0, Not Upgrading: 2123  
  Download size: 9388 kB / 9804 kB  
  Space needed: 63.7 MB / 789 MB available  
  
Continue? [Y/n] y  
Get:1 http://kali.download/kali kali-rolling/main amd64 gsasl-common all 2.2.2-1 [52.6 kB]  
Get:2 http://http.kali.org/kali kali-rolling/main amd64 guile-3.0-libs amd64 3.0.10+really3.0.10-4 [6877 kB]  
Get:4 http://mirror.aktkn.sg/kali kali-rolling/main amd64 libntlm0 amd64 1.8-4 [22.4 kB]  
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libgssglue1 amd64 0.9-1+b1 [20.8 kB]  
Get:5 http://kali.download/kali kali-rolling/main amd64 libgsasl18 amd64 2.2.2-1 [80.0 kB]  
Get:6 http://kali.download/kali kali-rolling/main amd64 mailutils-common all 1:3.19-1 [818 kB]  
Get:7 http://kali.download/kali kali-rolling/main amd64 libmailutils9t64 amd64 1:3.19-1 [943 kB]  
Get:8 http://kali.download/kali kali-rolling/main amd64 mailutils amd64 1:3.19-1 [574 kB]  
Fetched 9388 kB in 1min 7s (140 kB/s)  
Selecting previously unselected package gsasl-common.  
(Reading database ... 397051 files and directories currently installed.)  
Preparing to unpack .../0-gsasl-common_2.2.2-1_all.deb ...  
Unpacking gsasl-common (2.2.2-1) ...  
Selecting previously unselected package guile-3.0-libs:amd64.
```

Send a test email

```
(himara@vbox)-[~]  
$ echo "This is a test email" | mail -s "Test Email" vbox.kali
```

4.Linux GDB

1. Select the Appropriate Executable

Check the system architecture.

```
(himara@vbox)-[~]  
$ uname -m  
  
x86_64
```

- Based on the output, select the correct executable:
- If the output is x86_64, use the 64-bit executable.
- If the output is i686, use the 32-bit executable.
- If the output is arm, use the ARM-based executable.

2. Install the GDB



```
himara@vbox: ~/Desktop/Executables  
File Actions Edit View Help  
  
(himara@vbox)-[~/Desktop/Executables]  
$ sudo apt install gdb  
Upgrading:  
gdb  
  
Summary:  
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 2122  
  Download size: 3992 kB  
  Space needed: 0 B / 721 MB available  
  
Get:1 http://kali.download/kali kali-rolling/main amd64 gdb amd64 16.3-1 [3992 kB]  
Fetched 3992 kB in 3s (1566 kB/s)
```

3. Execute the file

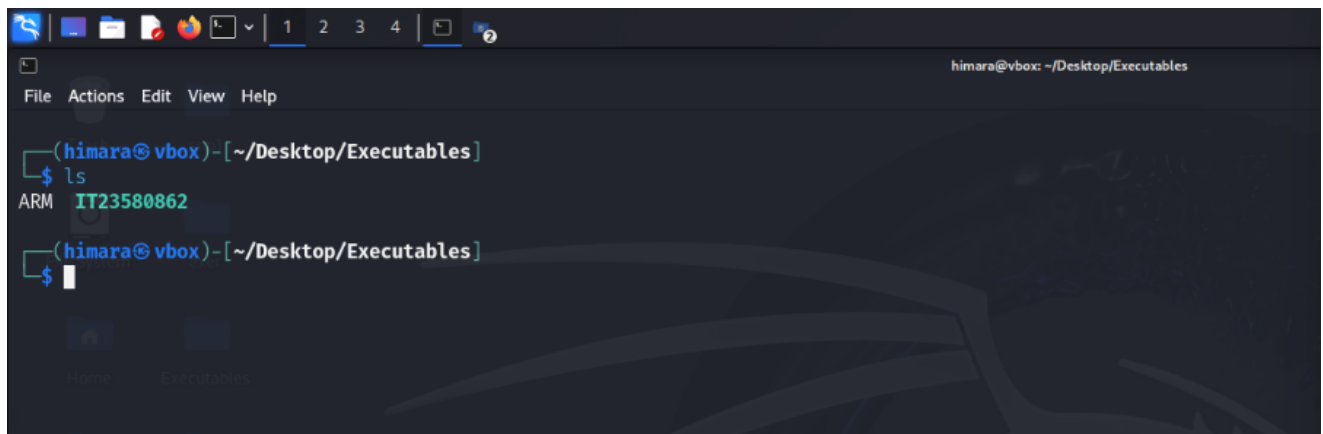


A terminal window titled 'himara@vbox: ~/Desktop' showing the process of unzipping a file. The user enters the command 'unzip Executables.zip'. The output shows the creation of the 'Executables/' directory and the inflation of three files: 'Executables/ARM', 'Executables/._ARM', and 'Executables/x86_64'. The terminal prompt returns to '(himara@vbox)-[~/Desktop]'.

```
(himara@vbox)-[~/Desktop]
$ unzip Executables.zip
Archive: Executables.zip
  creating: Executables/
  inflating: Executables/ARM
  inflating: __MACOSX/Executables/._ARM
  inflating: Executables/x86_64
  inflating: __MACOSX/Executables/._x86_64

(himara@vbox)-[~/Desktop]
$
```

4. Change the file



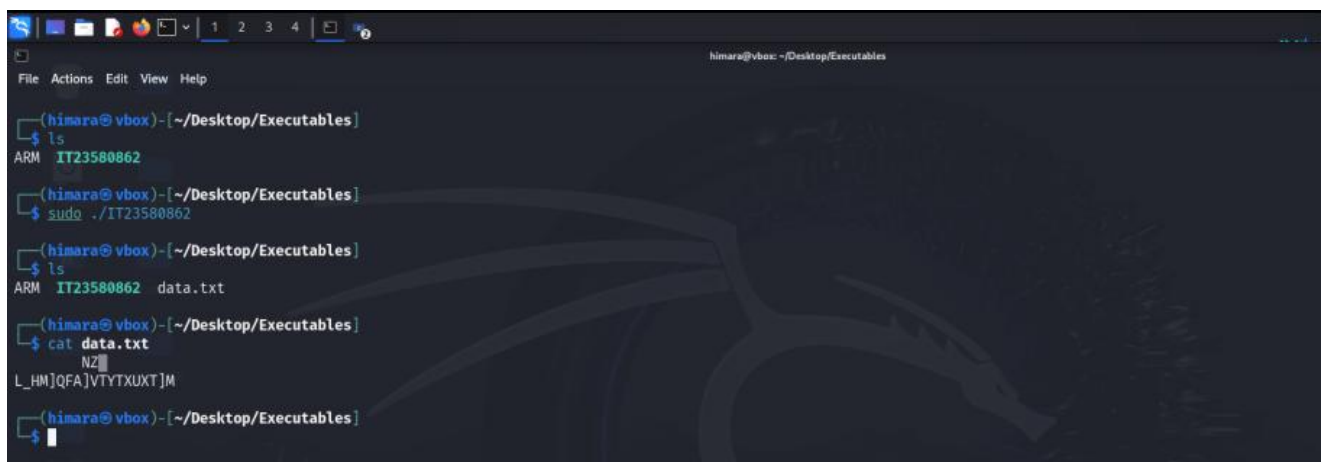
A terminal window titled 'himara@vbox: ~/Desktop/Executables' showing the contents of the directory. The user enters the command 'ls'. The output shows 'ARM' and 'IT23580862'. The terminal prompt returns to '(himara@vbox)-[~/Desktop/Executables]'.

```
(himara@vbox)-[~/Desktop/Executables]
$ ls
ARM  IT23580862

(himara@vbox)-[~/Desktop/Executables]
$
```

5. Analyze the generated executable

Execute the newly generated executable and analyze the output.



A terminal window titled 'himara@vbox: ~/Desktop/Executables' showing the execution and analysis of the generated executable. The user enters the command 'ls', then 'sudo ./IT23580862'. The output shows 'ARM IT23580862 data.txt'. The user then enters the command 'cat data.txt', and the output shows 'NZ' and 'L_HM]QFA]VTYTXUXT]M'. The terminal prompt returns to '(himara@vbox)-[~/Desktop/Executables]'.

```
(himara@vbox)-[~/Desktop/Executables]
$ ls
ARM  IT23580862

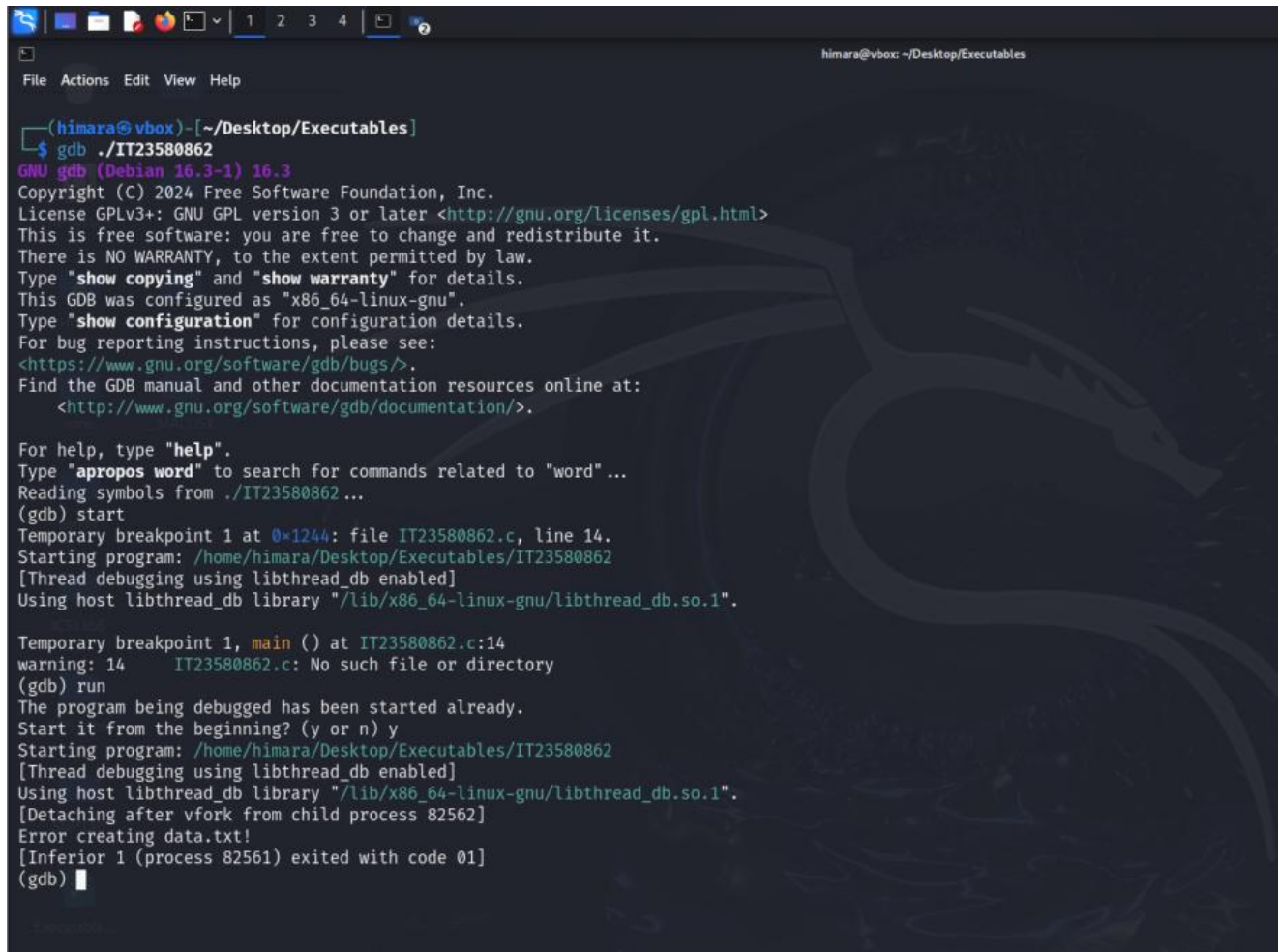
(himara@vbox)-[~/Desktop/Executables]
$ sudo ./IT23580862
ARM IT23580862 data.txt

(himara@vbox)-[~/Desktop/Executables]
$ cat data.txt
NZ
L_HM]QFA]VTYTXUXT]M

(himara@vbox)-[~/Desktop/Executables]
$
```


6. Debug the New Executable Using GDB

Start the debugging analyze the program flow.

A screenshot of a terminal window with a dark background and a faint dragon watermark. The terminal shows a GDB session. The user is in a directory ~/Desktop/Executables and runs 'gdb ./IT23580862'. GDB version 16.3 is shown. The user enters 'start', and GDB sets a temporary breakpoint at 0x1244 in file IT23580862.c, line 14. The program is started, and GDB shows it is using libthread_db. The program exits with code 01. The user then enters 'run', and GDB shows the program has already started. The user enters 'y' to start from the beginning, and GDB starts the program again. The program exits with code 01. The user enters 'q' to quit GDB.

```
(himara@vbox)-[~/Desktop/Executables]
$ gdb ./IT23580862
GNU gdb (Debian 16.3-1) 16.3
Copyright (C) 2024 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./IT23580862...
(gdb) start
Temporary breakpoint 1 at 0x1244: file IT23580862.c, line 14.
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Temporary breakpoint 1, main () at IT23580862.c:14
warning: 14      IT23580862.c: No such file or directory
(gdb) run
The program being debugged has been started already.
Start it from the beginning? (y or n) y
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[Detaching after vfork from child process 82562]
Error creating data.txt!
[Inferior 1 (process 82561) exited with code 01]
(gdb) q
```

```
File Actions Edit View Help
(gdb) run
The program being debugged has been started already.
Start it from the beginning? (y or n) y
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[Detaching after vfork from child process 82562]
Error creating data.txt!
[Inferior 1 (process 82561) exited with code 01]
(gdb) next
The program is not being run.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[Detaching after vfork from child process 82875]
Error creating data.txt!
[Inferior 1 (process 82874) exited with code 01]
(gdb) next
The program is not being run.
(gdb) break main
Breakpoint 2 at 0x55555555244: file IT23580862.c, line 14.
(gdb) next
The program is not being run.
(gdb) next
The program is not being run.
(gdb) break main
Note: breakpoint 2 also set at pc 0x55555555244.
Breakpoint 3 at 0x55555555244: file IT23580862.c, line 14.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 2, main () at IT23580862.c:14
14   in IT23580862.c
(gdb) next
[Detaching after vfork from child process 83056]
15   in IT23580862.c
(gdb) █
```

```
File Actions Edit View Help
The program is not being run.
(gdb) break main
Breakpoint 2 at 0x55555555244: file IT23580862.c, line 14.
(gdb) next
The program is not being run.
(gdb) next
The program is not being run.
(gdb) break main
Note: breakpoint 2 also set at pc 0x55555555244.
Breakpoint 3 at 0x55555555244: file IT23580862.c, line 14.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 2, main () at IT23580862.c:14
14   in IT23580862.c
(gdb) next
[Detaching after vfork from child process 83056]
15   in IT23580862.c
(gdb) next
20   in IT23580862.c
(gdb) next
21   in IT23580862.c
(gdb) next
23   in IT23580862.c
(gdb) next
25   in IT23580862.c
(gdb) next
26   in IT23580862.c
(gdb) next
27   in IT23580862.c
(gdb) next
28   in IT23580862.c
(gdb) next
29   in IT23580862.c
(gdb) next
Error creating data.txt!
30   in IT23580862.c
(gdb) █
```

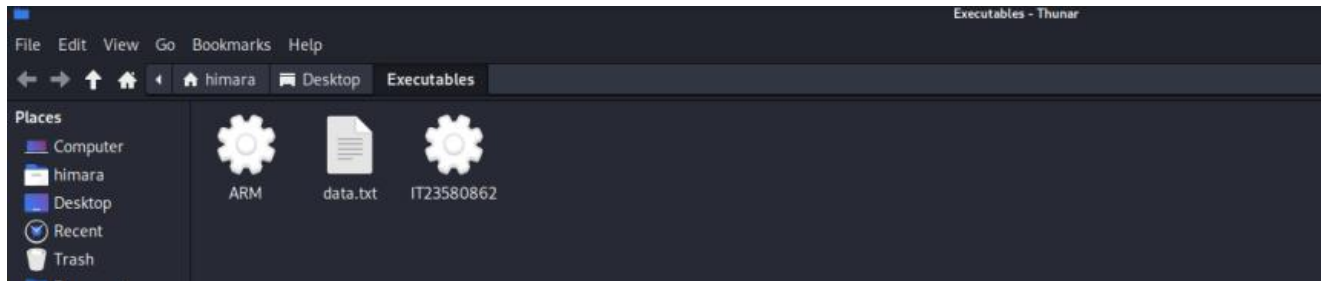
```
himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
Error creating data.txt!
30      in IT23580862.c
(gdb) break main
Note: breakpoints 2 and 3 also set at pc 0x55555555244.
Breakpoint 4 at 0x55555555244: file IT23580862.c, line 14.
(gdb) step
36      in IT23580862.c
(gdb) step
[Inferior 1 (process 83039) exited with code 01]
(gdb) step
The program is not being run.
(gdb) break main
Note: breakpoints 2, 3 and 4 also set at pc 0x55555555244.
Breakpoint 5 at 0x55555555244: file IT23580862.c, line 14.
(gdb) step
The program is not being run.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 2, main () at IT23580862.c:14
14      in IT23580862.c
(gdb) step
[Detaching after vfork from child process 83875]
15      in IT23580862.c
(gdb) step
20      in IT23580862.c
(gdb) step
21      in IT23580862.c
(gdb) step
s23      in IT23580862.c
(gdb) step
25      in IT23580862.c
(gdb) step
26      in IT23580862.c
(gdb) step
xor_encrypt_decrypt (data=0x7fffffffdd20 "ba71aaa2-ff54-464e-b863-21d30e31f6c4", key=0x555555556050 "key") at IT23580862.c:6
6      in IT23580862.c
(gdb) █
```

```
himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
(gdb) step
The program is not being run.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 2, main () at IT23580862.c:14
14      in IT23580862.c
(gdb) step
[Detaching after vfork from child process 83875]
15      in IT23580862.c
(gdb) step
20      in IT23580862.c
(gdb) step
21      in IT23580862.c
(gdb) step
s23      in IT23580862.c
(gdb) step
25      in IT23580862.c
(gdb) step
26      in IT23580862.c
(gdb) step
xor_encrypt_decrypt (data=0x7fffffffdd20 "ba71aaa2-ff54-464e-b863-21d30e31f6c4", key=0x555555556050 "key") at IT23580862.c:6
6      in IT23580862.c
(gdb) step
7      in IT23580862.c
(gdb) step
8      in IT23580862.c
(gdb) step
9      in IT23580862.c
(gdb) step
8      in IT23580862.c
(gdb) step
9      in IT23580862.c
(gdb) step
8      in IT23580862.c
(gdb) step
9      in IT23580862.c
(gdb) █
```

7. Flie system changes when debugging



8. Convert the executable file to a readable format

- Use the command: `strings ./IT23580862`. It will show the source code

```
(himara@vbox)-[~/Desktop/Executables]
$ strings ./IT23580862
/lib64/ld-linux-x86-64.so.2
__cxa_finalize
fgets
strncpy
__libc_start_main
fputs
fopen
fclose
pclose
strlen
popen
libc.so.6
GLIBC_2.2.5
GLIBC_2.34
_ITM_deregisterTMCloneTable
__gmon_start__
_ITM_registerTMCloneTable
PTE1
u+UH
sudo cat /sys/class/dmi/id/product_uuid
Error retrieving data
data.txt
Error creating data.txt!
;*3$"
GCC: (Debian 13.3.0-3) 13.3.0
__off_t
_IO_read_ptr
__chain
size_t
__shortbuf
_IO_buf_base
long long unsigned int
GNU C17 13.3.0 -mtune=generic -march=x86-64 -g -fasynchronous-unwind-tables
long long int
_fileno
_IO_read_end
_flags
```

```
himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
_freeres_list
main step
_IO_write_base vfork from child process 83875]
IT23580862.c
/home/himara/Desktop/Executables
/usr/lib/gcc/x86_64-linux-gnu/13/include
/usr/include/x86_64-linux-gnu/bits
/usr/include/x86_64-linux-gnu/bits/types
/usr/include
stddef.h
types.h
struct_FILE.h
stdio.h
string.h
#include <stdio.h>
#include <stdlib.h> (data=0x7fffffffdd20 "ba71aaa2-ff54-464e-b863-21d30e31f6c4", key=0x555555)
#include <string.h>
void xor_encrypt_decrypt(char *data, const char *key) {
7   size_t data_len = strlen(data);
(gdb) size_t key_len = strlen(key);
8   for (size_t i = 0; i < data_len; i++) {
(gdb) s data[i] ^= key[i % key_len];
9   }
int main() {
8   FILE *fp = popen("sudo cat /sys/class/dmi/id/product_uuid", "r");
(gdb) if (fp == NULL) {
9       printf("Error retrieving data\n");
(gdb) s return 1;
8   }
(gdb) char uuid[50];
9   fgets(uuid, sizeof(uuid), fp);
(gdb) fclose(fp);
8   uuid[strcspn(uuid, "\n")] = 0; // Remove newline
(gdb) const char *key = "key";
9   xor_encrypt_decrypt(uuid, key);
(gdb) FILE *out = fopen("data.txt", "w");
8   if (out == NULL) {
(gdb) s printf("Error creating data.txt!\n");
9       return 1;
(gdb) }
```



```

23 if (out == NULL) {
(gdb) s printf("Error creating data.txt!\n");
25     return 1;
(gdb) step
26 fprintf(out, "%s", uuid);
(gdb) fclose(out);
27     return 0;
Scrt1.o
__abi_tag
crtstuff.c
deregister_tm_clones
__do_global_dtors_aux
completed.0
__do_global_dtors_aux_fini_array_entry
frame_dummy
__frame_dummy_init_array_entry
IT23580862.c
__FRAME_END__
_DYNAMIC
__GNU_EH_FRAME_HDR
__GLOBAL_OFFSET_TABLE__
__libc_start_main@GLIBC_2.34
_ITM_deregisterTMCloneTable
edata
fclose@GLIBC_2.2.5
_fini
strlen@GLIBC_2.2.5
pclose@GLIBC_2.2.5
fputs@GLIBC_2.2.5
strcspn@GLIBC_2.2.5
fgets@GLIBC_2.2.5

```

- Copy the C code and create the file using this command nano SNP.c and paste the code.

```

himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
GNU nano 8.1 SNP.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
void xor_encrypt_decrypt(char *data, const char *key) {
    size_t data_len = strlen(data);
    size_t key_len = strlen(key);
    for (size_t i = 0; i < data_len; i++) {
        data[i] ^= key[i % key_len];
    }
}
int main() {
    FILE *fp = popen("sudo cat /sys/class/dmi/id/product_uuid", "r");
    if (fp == NULL) {
        printf("Error retrieving data\n");
        return 1;
    }
    char uuid[50];
    fgets(uuid, sizeof(uuid), fp);
    pclose(fp);
    uuid[strcspn(uuid, "\n")] = 0; // Remove newline
    const char *key = "key";
    xor_encrypt_decrypt(uuid, key);
    FILE *out = fopen("data.txt", "w");
    if (out == NULL) {
        printf("Error creating data.txt!\n");
        return 1;
    }
    fprintf(out, "%s", uuid);
    fclose(out);
    return 0;
}

```

Run the source code

```

(himara@vbox)~[~/Desktop/Executables]
$ nano SNP.c
(himara@vbox)~[~/Desktop/Executables]
$ sudo chmod +x SNP.c
(himara@vbox)~[~/Desktop/Executables]
$ sudo gcc SNP.c -o snp
(himara@vbox)~[~/Desktop/Executables]
$

```

9. Create a Python script to decode the data.txt

```
himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
GNU nano 8.1 decrypt.py *
def xor_decrypt(data, key):
    decrypted = ''
    for i in range(len(data)):
        decrypted += chr(ord(data[i]) ^ ord(key[i % len(key)]))
    return decrypted
#read the encrypted data
with open('data.txt', 'r') as f:
    encrypted = f.read()
key = "key"
decrypted_uuid = xor_decrypt(encrypted, key)
print("Decrypted UUID:", decrypted_uuid)
```

- Using this command **python3 decrypt.py** you can decrypt the data.txt.

```
(himara@vbox)-[~/Desktop/Executables]
$ nano decrypt.py
(gdb) next
(himara@vbox)-[~/Desktop/Executables]
$ python3 decrypt.py
Decrypted UUID: ba71aaa2-af54-464e-b863-21d30e31f6c4
(gdb) step
(himara@vbox)-[~/Desktop/Executables]
$
```

- During GDB debugging, we discovered that the source code has XOR encryption. After conducting additional string analysis, we discovered the entire code, transferred it to a C file, and executed it.
- We demonstrated that it was responsible for creating the data.txt.
- A custom Python code was used to decrypt Data.txt, and the result displayed the system UUID, which is the encrypted content.